

ESPRIMO G6012 / G9012 / G9013

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System Upgrades and Repairs



We recommend printing out the relevant sections of this chapter prior to installation as the device for installing/removing hardware components needs to be switched off.

The following illustrations may differ from your device, depending on its configuration and features.

If you have received any further documents with your device, you should read these through carefully.

Please note the following instructions before installing/removing system components:



- The device must be switched off when installing/removing components and it must not be in power-saving mode.
- Disconnect the mains plug before opening the device.
- Discharge the remaining voltage by pressing the power switch for several seconds.
- Remove all externally connected cables.
- Take care not to pinch any cables when installing/removing components.
- Please ensure that the installed components that might heat up intensely during operation do not exceed the permitted operating temperature.

Please take note of the target audience addressed in this manual. Damage to the device incurred as a result of unqualified methods used during repair work as well as the use of unsuitable spare parts or assemblies during the repair work or upgrades of the device may void your warranty claim.



During a system expansion or a hardware upgrade, it is possible that the BIOS must be updated. You can find more information in the technical information provided for your device.

Information on the installation and removal of components

Please pay attention to the locking mechanism (indentations and centring pins) when installing hardware components.

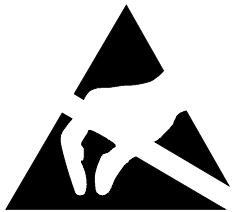
Note that some components on the mainboard may be very hot if the device was in use shortly before the case was removed.

Utmost care is required during installation and removal to avoid damaging components.

Never use items with sharp edges (screwdrivers) to lever them out.

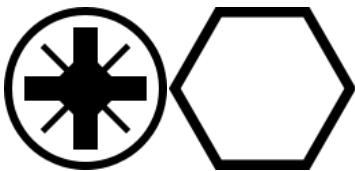
Parts with Electrostatic Sensitive Devices (ESD) can be identified by a corresponding label.

When dealing with boards with ESDs, the following points must always be observed:



- Always ensure there is a static discharge, e.g. by touching a grounded object prior to commencing with the work.
- The devices and tools being used must not be statically charged.
- Only touch the components along the edges or on the areas marked in green (contact points).
- Contacts and electronic components must not touch when handling components equipped with ESD protection.

Screws

		 M2,5 x 4,0
Screw: M2.5 x 4.0 Screwdriver: Phillips PH 1 (ISO 8764)		
		 M3 x 3,5
Screw: M3 x 3.5 Screwdriver: Phillips PH 1 (ISO 8764)		
		
Screw: Knurled thumb screw Screwdriver: Phillips PH 1 (not essential) (ISO 8764)		
		 M3 x 4,5
Screw: M3 x 4.5 Screwdriver: Hexagon-head size 5 (ISO 1711) Screwdriver: Pozidriv PZ 2 (ISO 8764)		
		 M2 x 3,0
Screw: M2 x 3.0 Screwdriver: Torx Plus 6IP (ISO 8764)		

		 M2 x 3,5
Screw: M2 x 3.5 Screwdriver: Y-Type (Tri-Point) Y1 (ISO 1173)		
		 M2,5 x 3,0
Screw: M2.5 x 3.0 Screwdriver: Phillips PH 1 (ISO 8764)		
		 M3 x 5,0
Screw: M3 x 5.0 Screwdriver: Phillips PH 1 (ISO 8764)		

Distinction between housing variants

Required work steps:
■ None

Required tools:
■ None

The ESPRIMO G6012 is available in two, the ESPRIMO G9012 and G9013 in 3 housing variants:

Housing small



Housing ODD



Housing internal PSU



Work steps that only pertain to one housing variant are marked accordingly.

Open the cable cover

Required work steps:

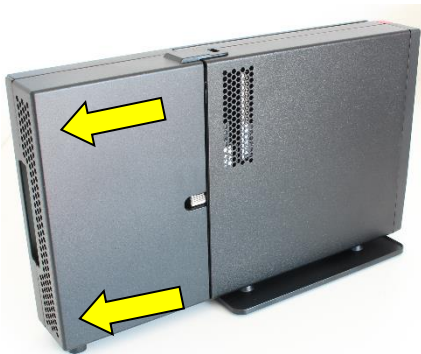
■ None

- ▶ Disconnect the mains plug from the mains outlet.
- ▶ Discharge the remaining voltage by pressing the power switch for several seconds.

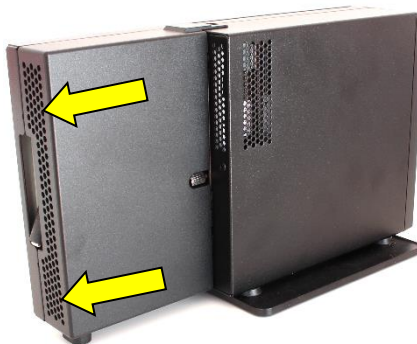
Required tools:

■ None

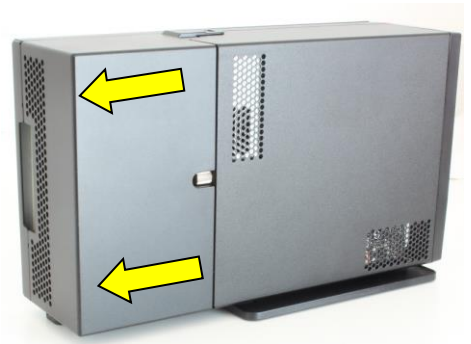
Housing small



Housing ODD



Housing internal PSU



- ▶ Pull the cable cover in the direction of the arrow and remove it.

System with opened cable cover:

Housing small



Housing ODD



Housing internal PSU



Installation:

- ▶ Proceed in reverse order.

Remove the cable cover

Required work steps:

- [Open the cable cover](#)

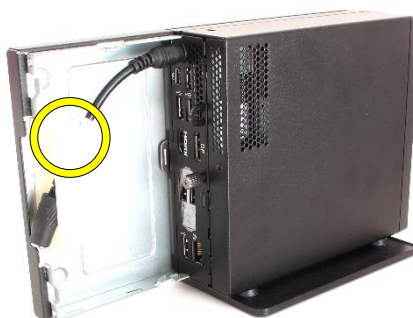
Required tools:

- Screwdriver: Phillips PH 1

Housing small



Housing ODD

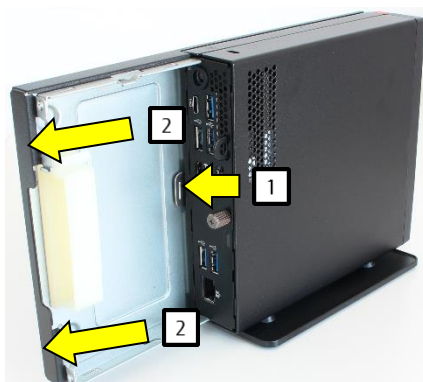


Housing internal PSU

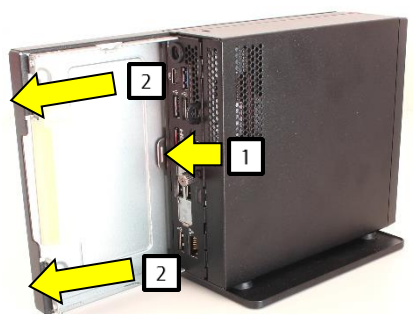


- Open the cable clip and remove all cables connected to the computer.

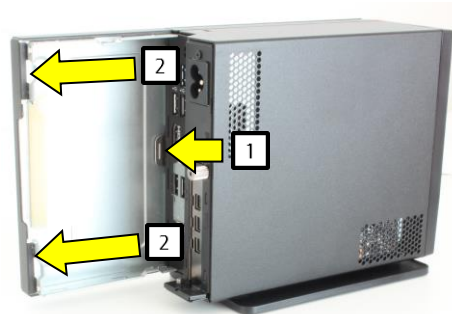
Housing small



Housing ODD



Housing internal PSU



- Push the tab (1), pull the cable cover in the direction of the arrow (2) and remove.



M2,5 x 4,0

- Remove the screws and detach the mounting plate.

Cable cover:

Housing small and Housing ODD

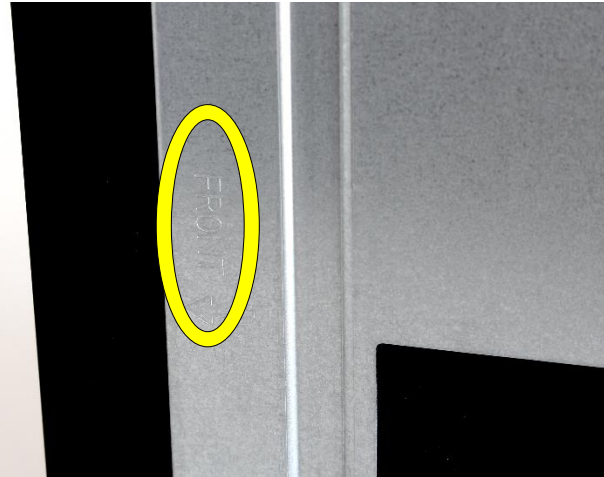


Housing internal PSU



Installation:

- ▶ Proceed in reverse order.



- ▶ Make sure the assembly plate is correctly aligned (check for FRONT marking) during installation.

Remove the pedestal

Required work steps:

■ None

- ▶ Disconnect the mains plug from the mains outlet.
- ▶ Remove all cables connected to the computer.
- ▶ Discharge the remaining voltage by pressing the power switch for several seconds.

Required tools:

■ Screwdriver: Phillips PH 1

Housing small



Housing ODD



Housing internal PSU



- ▶ Remove the screws.



M2,5 x 4,0

Pedestal:

Housing small



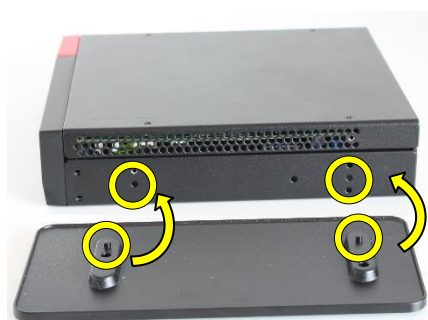
Housing ODD and Housing internal PSU



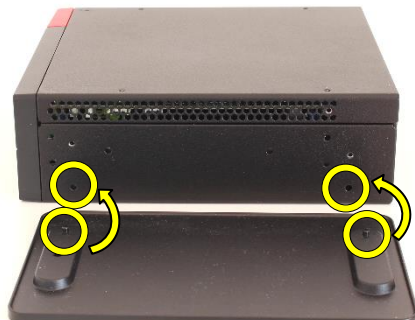
Installation:

- ▶ Proceed in reverse order.

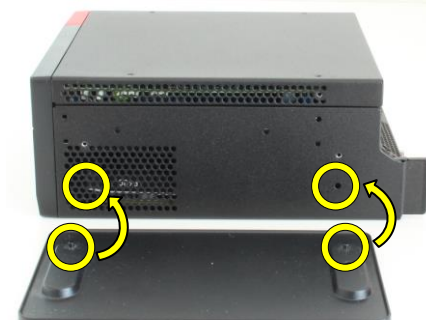
Housing small



Housing ODD



Housing internal PSU



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Remove the VESA mount

Required work steps:

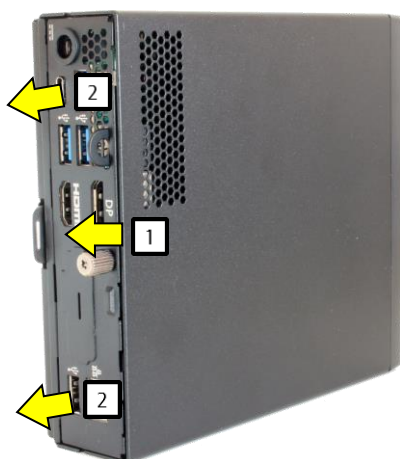
■ None

Required tools:

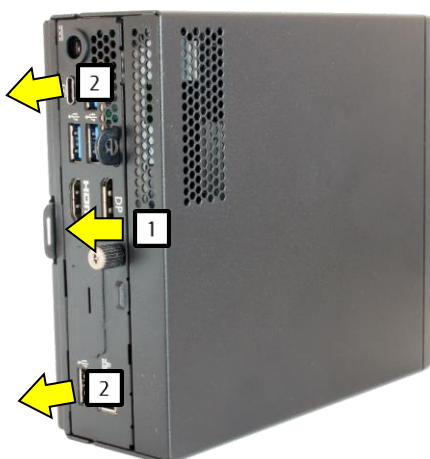
■ Screwdriver: Phillips PH 1

- ▶ Disconnect the mains plug from the mains outlet.
- ▶ Remove all cables connected to the computer.
- ▶ Discharge the remaining voltage by pressing the power switch for several seconds.

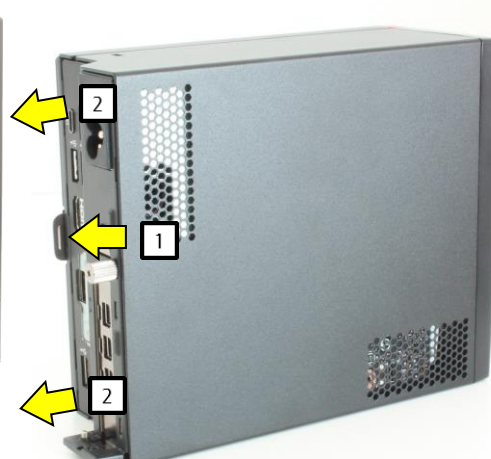
Housing small



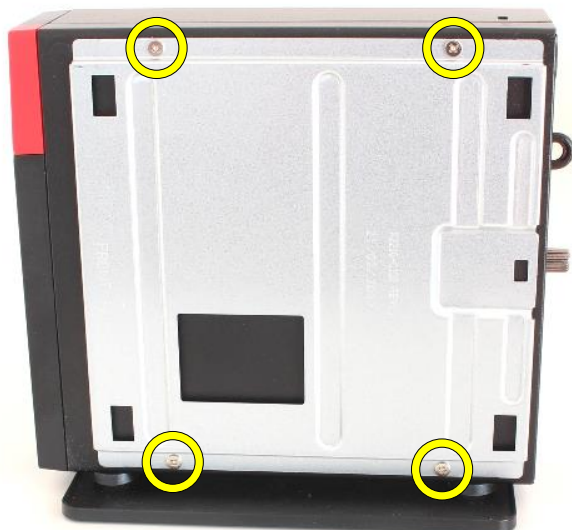
Housing ODD



Housing internal PSU



- ▶ Push the tab (1), pull the VESA mount in the direction of the arrow (2) and remove.



M2,5 x 4,0

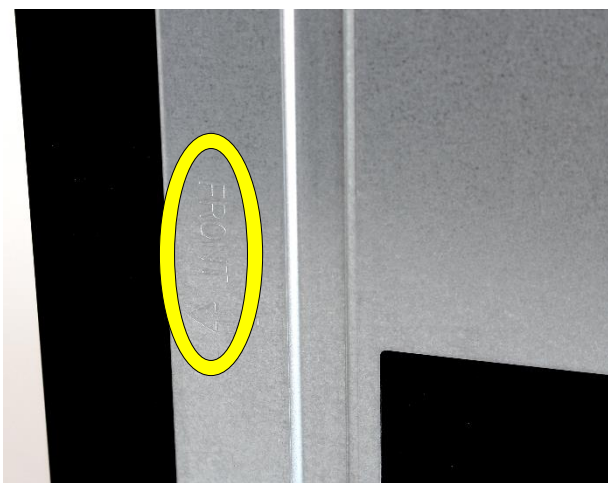
- ▶ Remove the screws and detach the mounting plate.

VESA mount:



Installation:

- ▶ Proceed in reverse order.



- ▶ Make sure the assembly plate is correctly aligned (check for FRONT marking) during installation.

Remove the dust filter

Required work steps:

■ None

Required tools:

■ None

- ▶ Disconnect the mains plug from the mains outlet.
- ▶ Remove all cables connected to the computer.
- ▶ Discharge the remaining voltage by pressing the power switch for several seconds.

Housing small

Housing ODD

Housing internal PSU



- ▶ Remove the dust filter.

Dust filter:



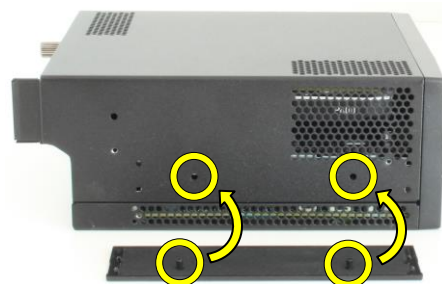
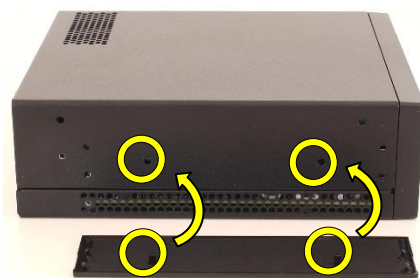
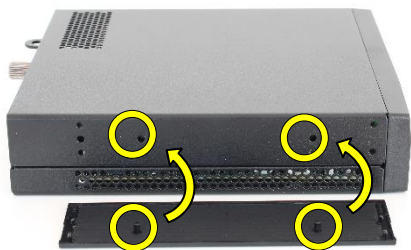
Installation:

- ▶ Proceed in reverse order.

Housing small

Housing ODD

Housing internal PSU



Open the case

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)

Required tools:

- None

- ▶ Disconnect the mains plug from the mains outlet.
- ▶ Remove all cables connected to the computer.
- ▶ Discharge the remaining voltage by pressing the power switch for several seconds.

Housing small



- ▶ Unscrew the knurled screw.

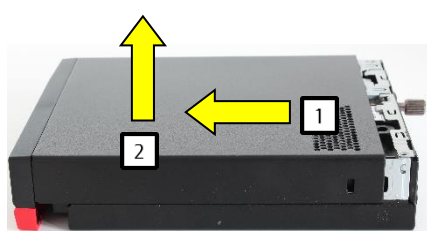
Housing ODD



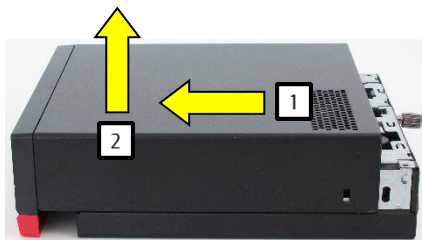
Housing internal PSU



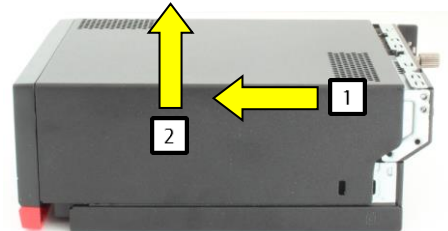
Housing small



Housing ODD



Housing internal PSU



- ▶ Slide the housing cover in the direction of the arrow (1) and remove (2).

System without housing cover:

Housing small



Housing ODD



Housing internal PSU



Installation:

- ▶ Proceed in reverse order.

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Remove the drive bay (Housing small)

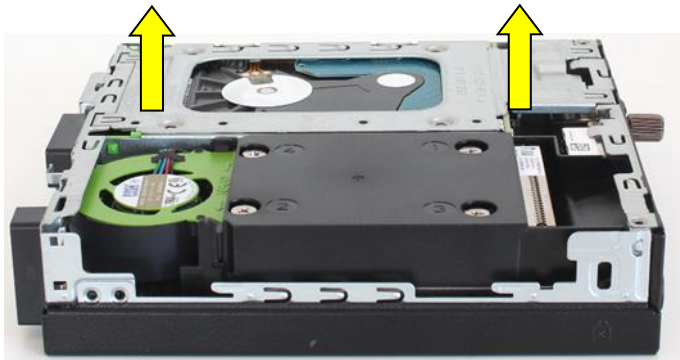
Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)

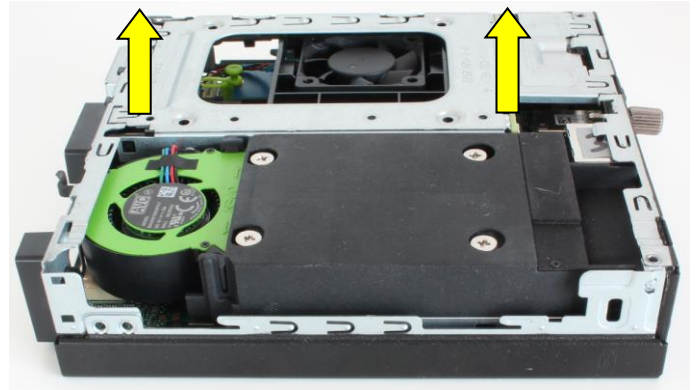
Required tools:

- None

ESPRIMO G9012 with HDD:



ESPRIMO G9013 with memory cooling fan:



- ▶ Pull the drive bay vertically upwards.
On the ESPRIMO G9013, make sure that the fan cable of the memory cooling fan is still plugged in.



- ▶ Disconnect the fan cable.

Drive bay:

ESPRIMO G9012 with HDD



ESPRIMO G9013 with memory cooling fan



Installation:

- ▶ Proceed in reverse order.

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Remove the second M.2 SSD module (Housing small)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)

Required tools:

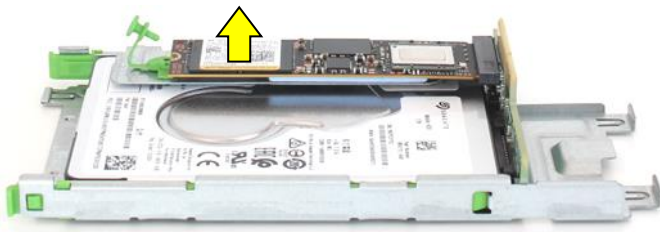
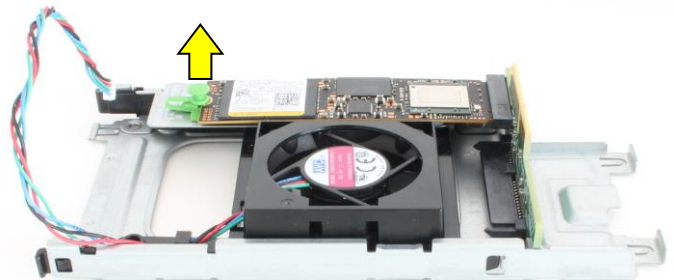
- None

ESPRIMO G9012 with HDD

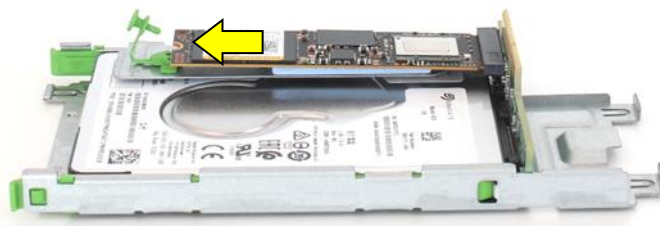
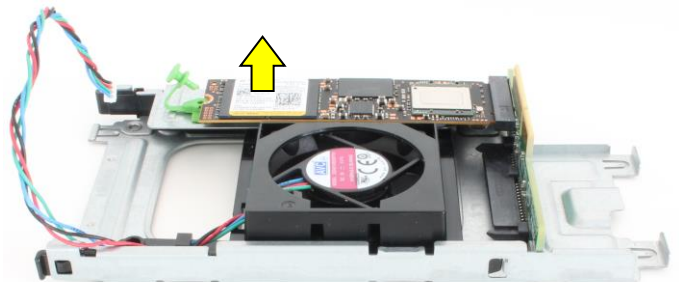


- ▶ Remove the pin from the connector lock.

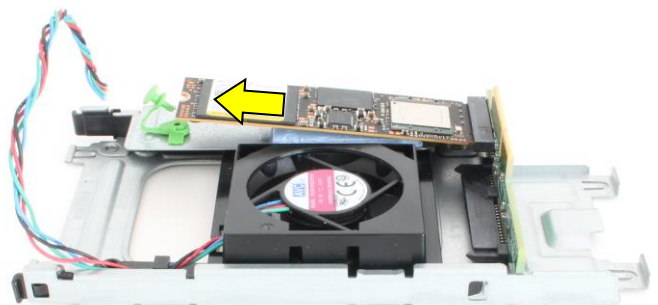
ESPRIMO G9013 with memory cooling fan



- ▶ Carefully lift the module until it disconnects from the thermal pad.



- ▶ Disconnect the module.



Installation:

- ▶ Proceed in reverse order.



- ▶ Make sure that a thermal pad is available.
- ▶ Make sure to align it properly when inserting the module.

Remove the 2.5-inch drive (Housing small)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)

Required tools:

- None



- Press the rails together (1) and disconnect the drive in the direction of the arrow (2).



- Remove the rails.

Installation:



- Attach the rails to the drive (insert pins into the threaded holes). Pay attention to the direction.
- Slide the drive into its bay until it snaps in place.

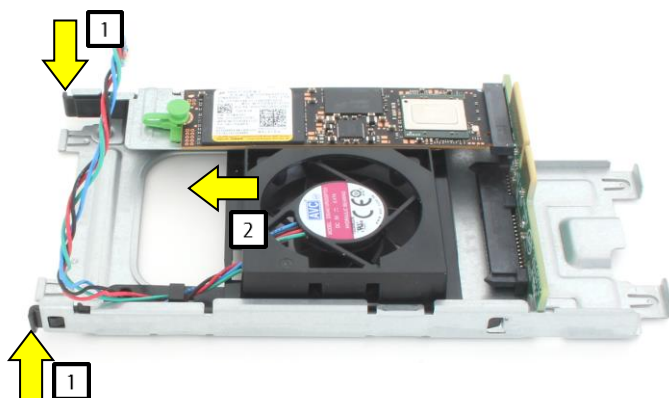
Remove the memory cooling fan (Housing small)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)

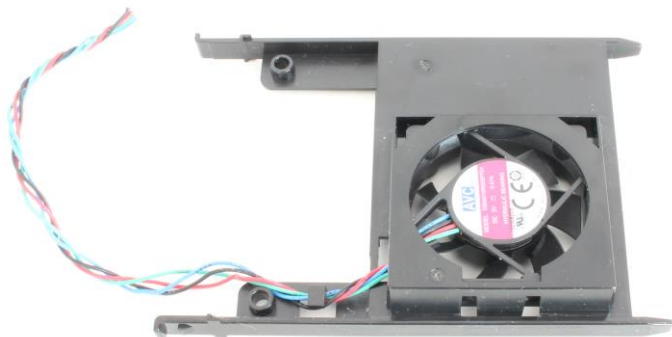
Required tools:

- None



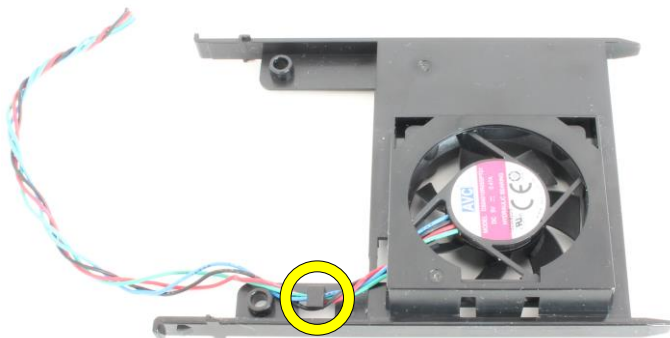
- ▶ Press the rails together (1) and pull out the fan module in the direction of the arrow (2).

Memory cooling fan:



Installation:

- ▶ Proceed in reverse order.



- ▶ Make sure that the fan cable is in the cable guide.

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Remove the Riser board (Housing small)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the second M.2 SSD module \(Housing small\)](#)
- [Remove the 2.5-inch drive \(Housing small\)](#)

Required tools:

- Screwdriver: Phillips PH 1



- ▶ Remove the screws.



Riser board



Drive bay without 2.5-inch drive



Drive bay with memory cooling fan



Riser board and drive bay:

Installation:

- ▶ Proceed in reverse order.



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Remove the drive bay (Housing ODD)

Required work steps:

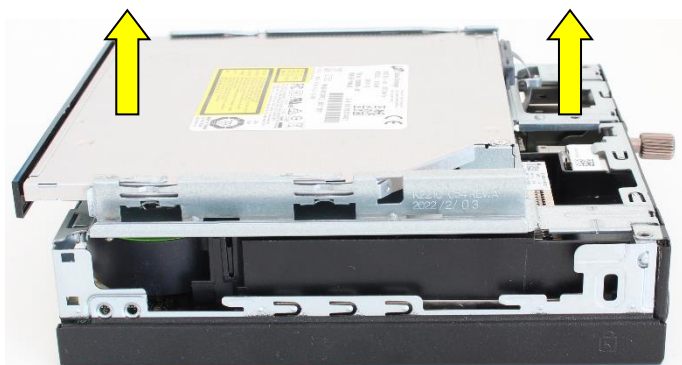
- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)

Required tools:

- None

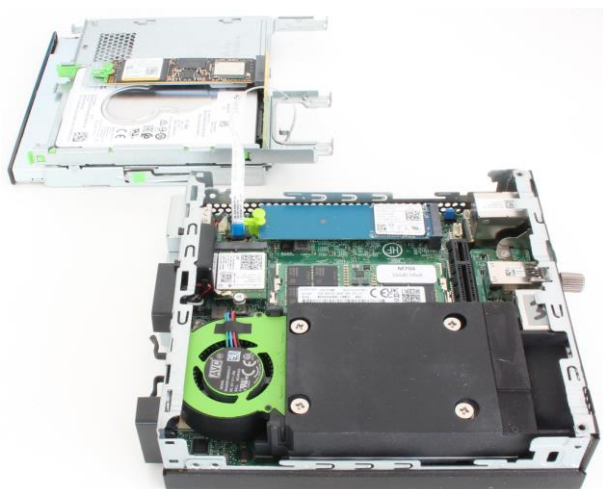


The cable connections of the fan and the visual drive to the mainboard are short. Proceed carefully with the next step.



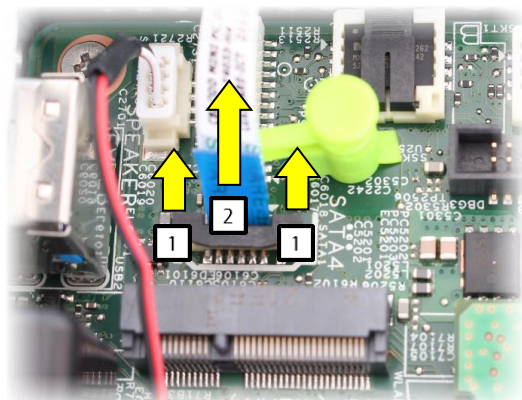
- Pull the drive bay vertically upwards.

ESPRIMO G9012 with HDD



- Place the drive bay next to the device.

ESPRIMO G9013 with memory cooling fan



- Disconnect the plug (1) and pull out the cable (2).

ESPRIMO G9013 with memory cooling fan



► Disconnect the fan cable.

Drive bay:

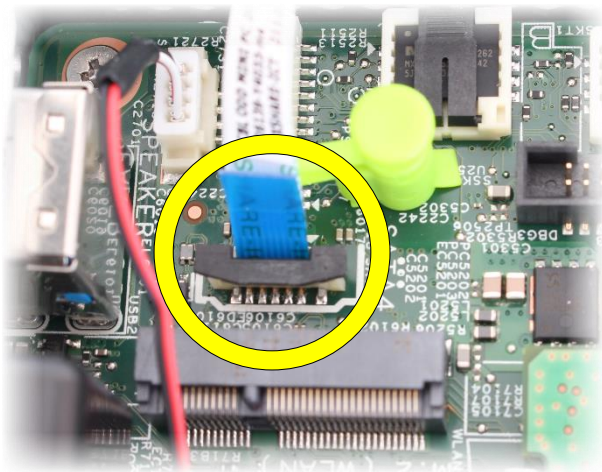
ESPRIMO G9012 with HDD



ESPRIMO G9013 with memory cooling fan



Installation:



► Proceed in reverse order.
Make sure the cable is aligned correctly.

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Remove the second M.2 SSD module (Housing ODD)

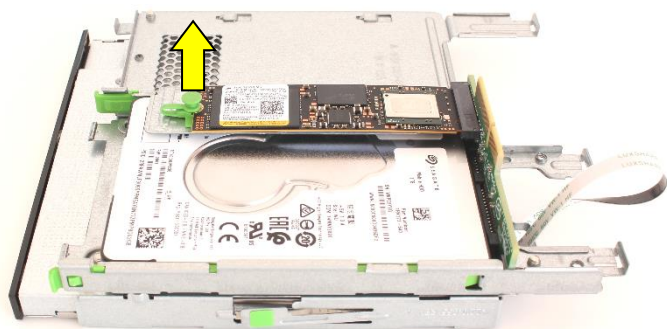
Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing ODD\)](#)

Required tools:

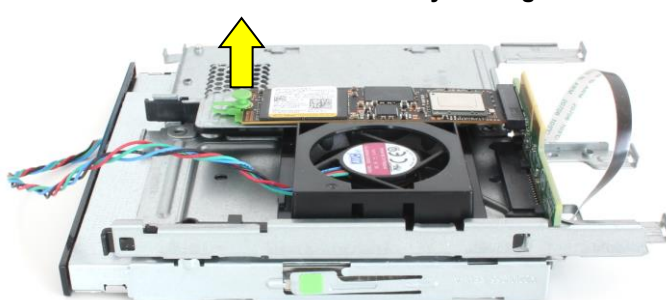
- None

ESPRIMO G9012 with HDD



- Remove the pin from the connector lock.

ESPRIMO G9013 with memory cooling fan



- Carefully lift the module until it disconnects from the thermal pad.



- Disconnect the module.



Installation:

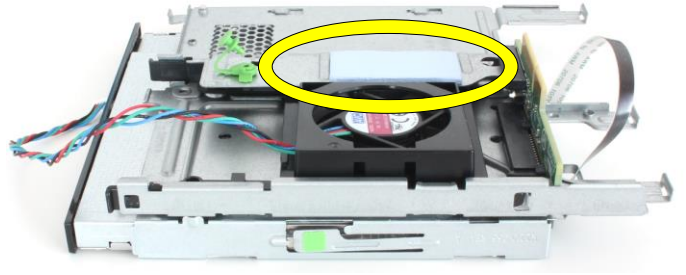
- ▶ Proceed in reverse order.

ESPRIMO G9012 with HDD



- ▶ Make sure that a thermal pad is available.
- ▶ Make sure to align it properly when inserting the module.

ESPRIMO G9013 with memory cooling fan



Remove the 2.5-inch drive (Housing ODD)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing ODD\)](#)

Required tools:

- None



- ▶ Press the rails together (1) and disconnect the drive in the direction of the arrow (2).



- ▶ Remove the rails.

Installation:



- ▶ Attach the rails to the drive (insert pins into the threaded holes). Pay attention to the direction.
- ▶ Slide the drive into its bay until it snaps in place.

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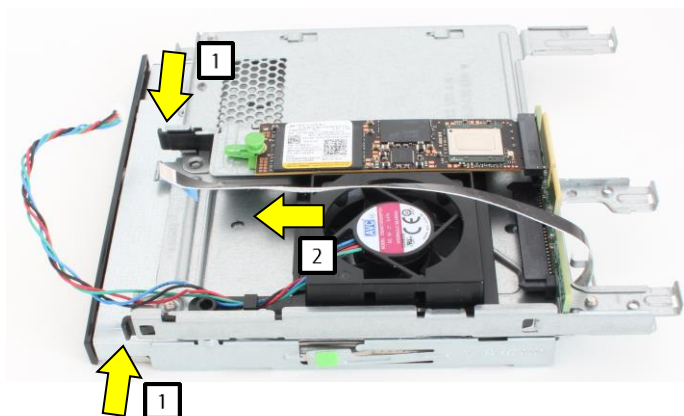
Remove the memory cooling fan (Housing ODD)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing ODD\)](#)

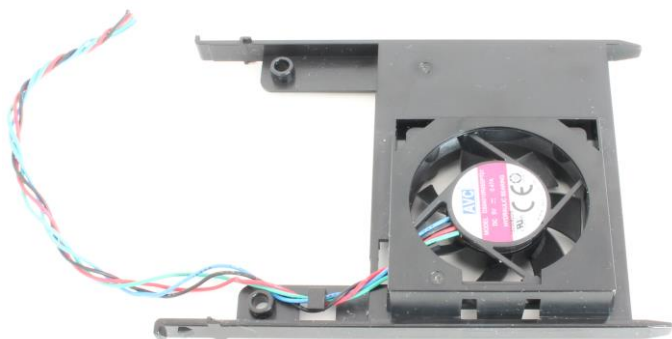
Required tools:

- None



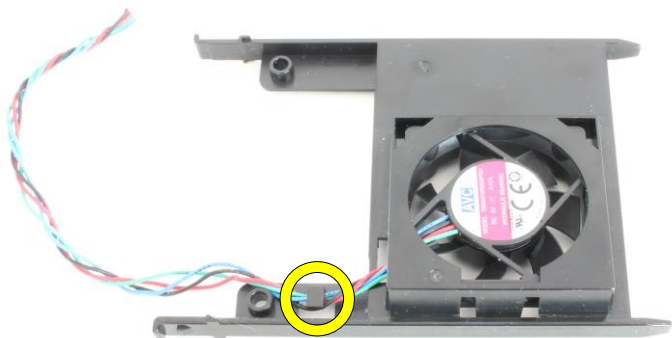
- ▶ Press the rails together (1) and disconnect the drive in the direction of the arrow (2).

Memory cooling fan:



Installation:

- ▶ Proceed in reverse order.



- ▶ Make sure that the fan cable is in the cable guide.

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Remove the optical disc drive (Housing ODD)

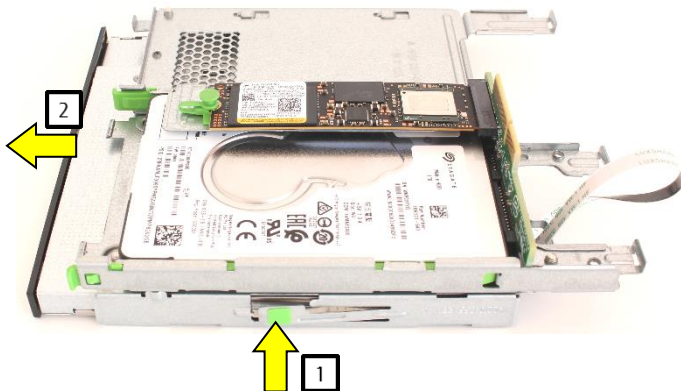
Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing ODD\)](#)

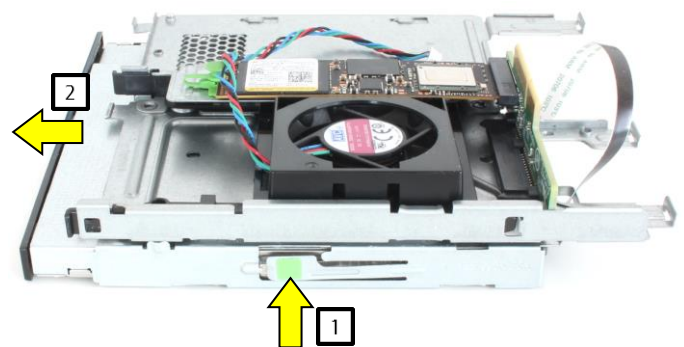
Required tools:

- None

ESPRIMO G9012 with HDD



ESPRIMO G9013 with memory cooling fan



- Press the lock release (1) and disconnect the drive in the direction of the arrow (2).



- Remove the rail.

Installation:



- ▶ Attach the rail to the drive (insert pins into the threaded holes).



- ▶ Pay attention to the installation direction.

ESPRIMO G9012 with HDD



ESPRIMO G9013 with memory cooling fan



- ▶ Slide the disc drive into the casing until the connector lock snaps into place.

Remove the Riser board (Housing ODD)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the second M.2 SSD module \(Housing ODD\)](#)
- [Remove the 2.5-inch drive \(Housing ODD\)](#)

Required tools:

- Screwdriver: Phillips PH 1

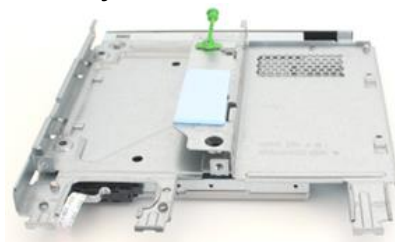


- ▶ Remove the screws.

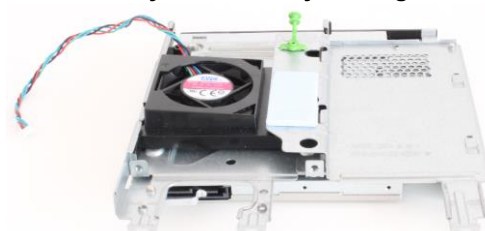
Riser board



Drive bay without 2.5-inch drive



Drive bay with memory cooling fan



Installation:

- ▶ Proceed in reverse order.



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Remove the PCIe card (Housing internal PSU)

Required work steps:

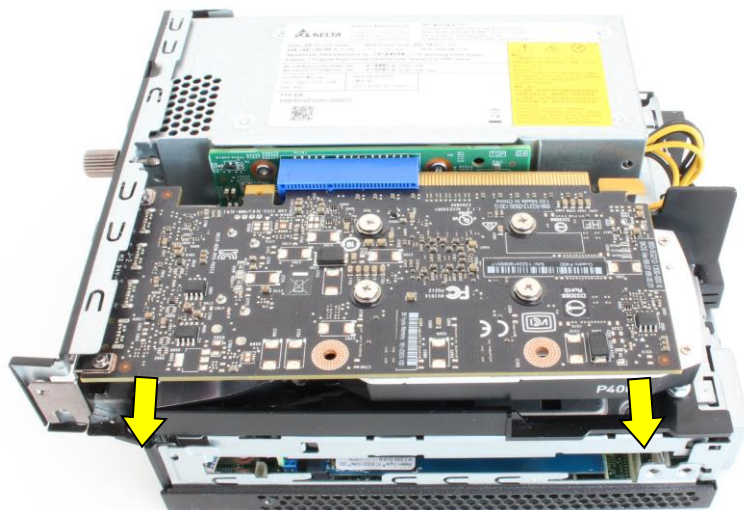
- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)

Required tools:

- Screwdriver: Phillips PH 1



- ▶ Remove the screw.



- ▶ Remove the PCIe card.

System without PCIe cards:



Installation:

- ▶ Proceed in reverse order.

Remove the power supply unit (Housing internal PSU)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)

Required tools:

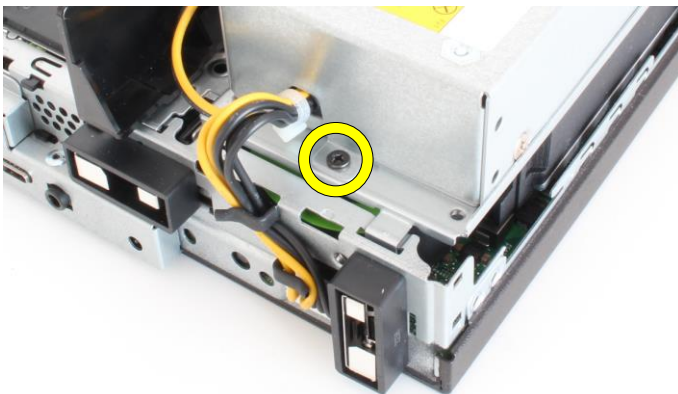
- Screwdriver: Phillips PH 1



- ▶ Remove the screws.



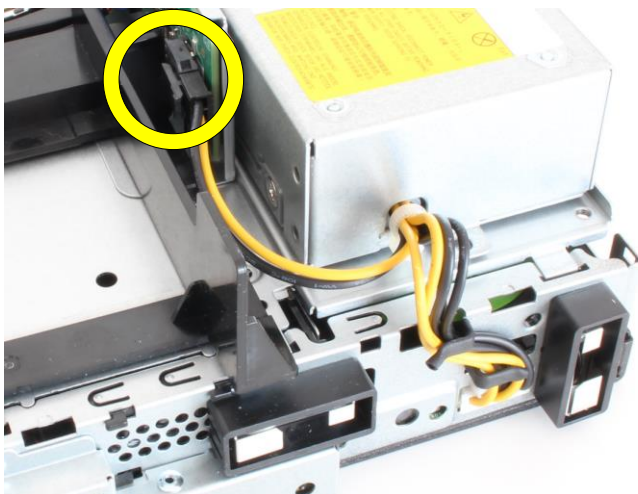
M2,5 x 3,0



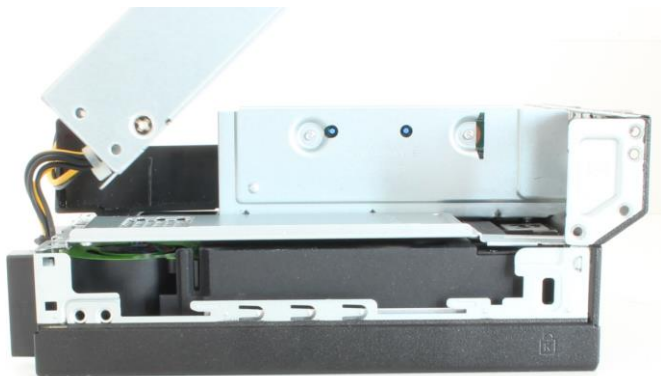
- ▶ Remove the screw.



M2,5 x 3,0



- ▶ Press the clip and disconnect the cable.



- ▶ Lift the power supply unit out.



- ▶ Remove the cable from the guides (1) and disconnect the plug (2).

Power supply unit



Installation:

- ▶ Proceed in reverse order.
Make sure that the cable is correctly clipped in the guides.

Remove the drive bay (Housing internal PSU)

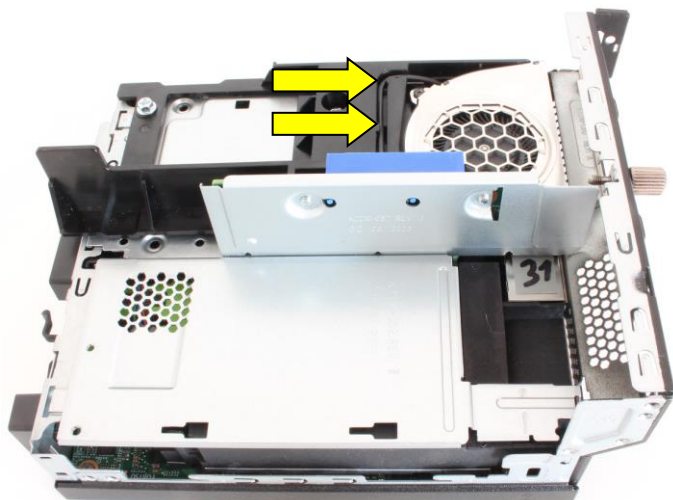
Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)

Required tools:

- None

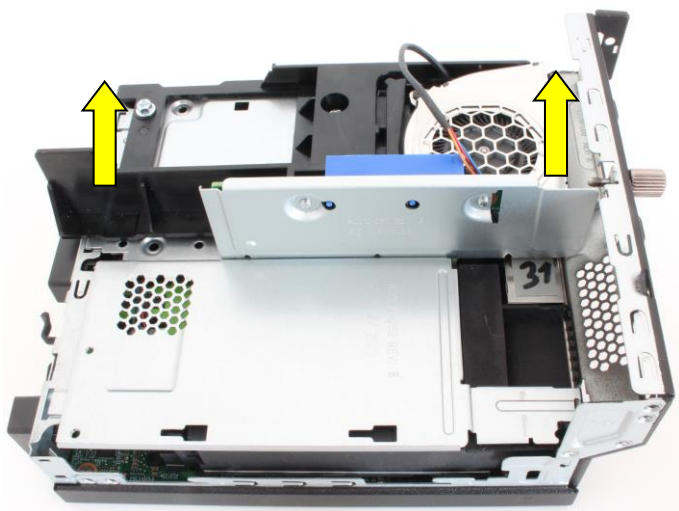
ESPRIMO G9012 or G9013 with PCIe fan:



- Remove the cable from the guide.



The cable connections of the fans and the hard disk to the mainboard are short. Proceed carefully with the next step.



- Pull the drive bay vertically upwards.

ESPRIMO G9012 with HDD



- ▶ Place the drive bay next to the device.

ESPRIMO G9013 with memory cooling fan

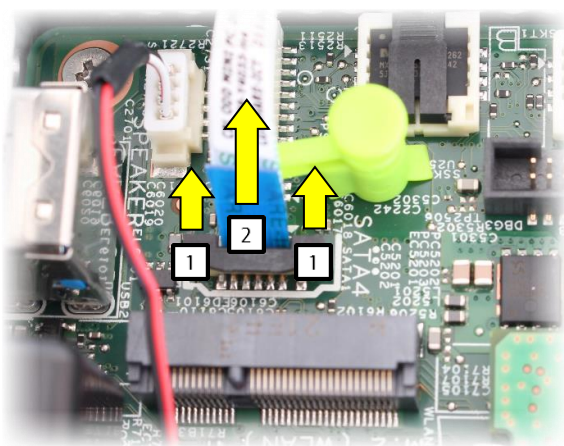


ESPRIMO G9012 or G9013 with PCIe fan:



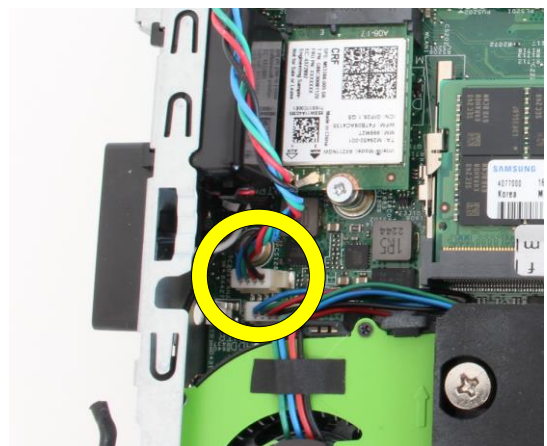
- ▶ Disconnect the cable.

ESPRIMO G9012 with HDD



- ▶ Disconnect the plug (1) and pull out the cable (2).

ESPRIMO G9013 with memory cooling fan



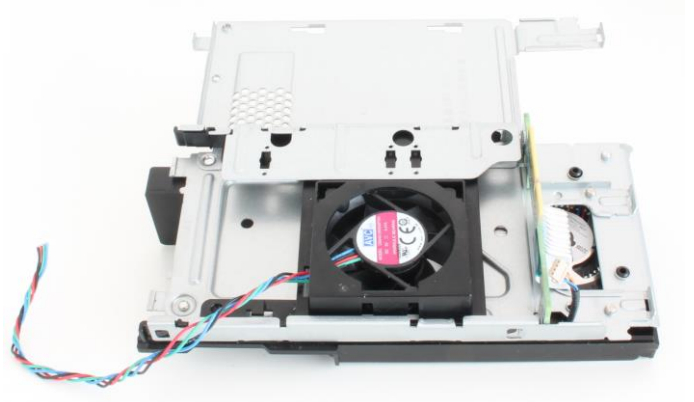
- ▶ Disconnect the fan cable.

Drive bay:

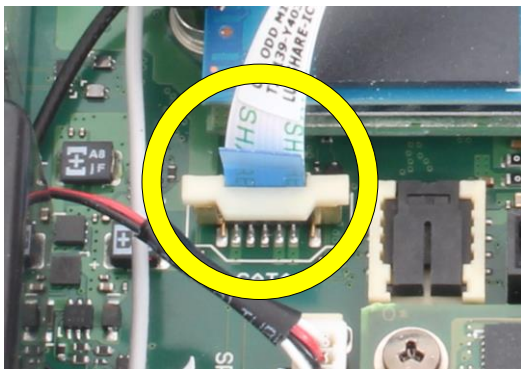
ESPRIMO G9012 with HDD



ESPRIMO G9013 with memory cooling fan



Installation:



- ▶ Proceed in reverse order.
Make sure the alignment is correct when inserting the plug.



When assembling, take care not to pinch any cables.

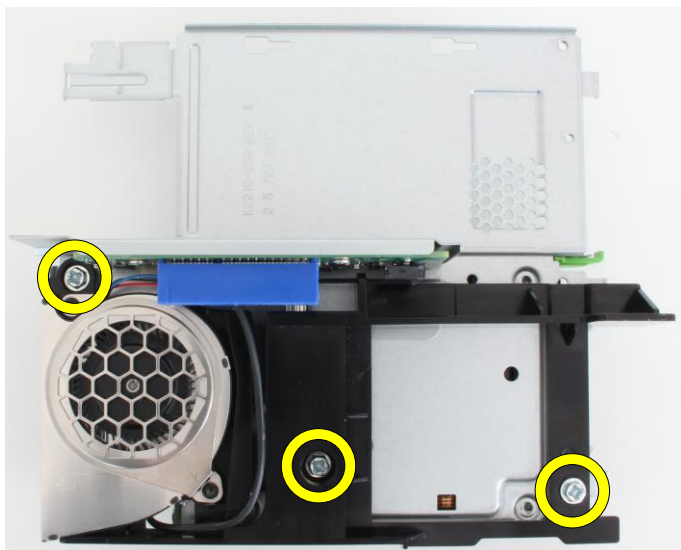
Remove the PCIe fan module (Housing internal PSU)

Required work steps:

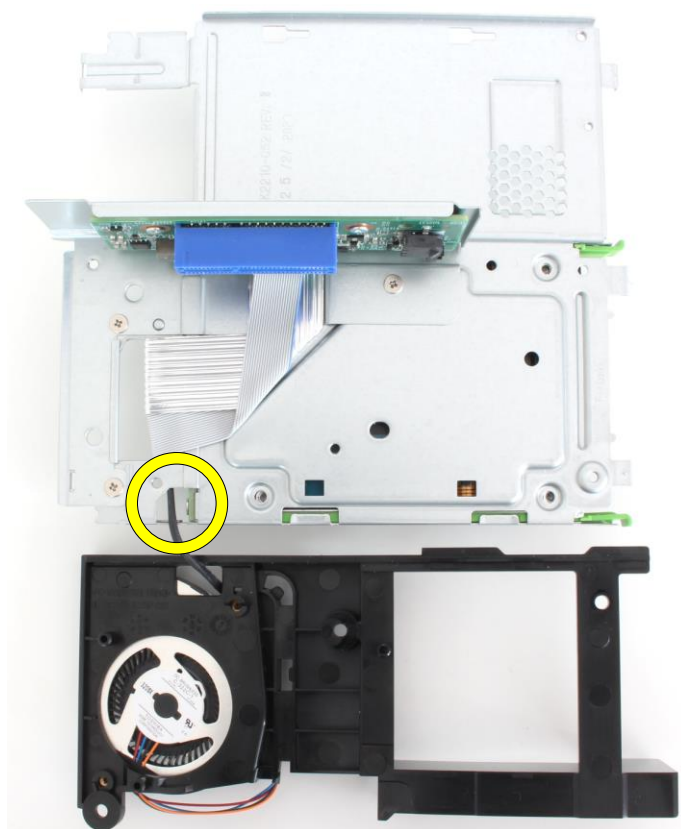
- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)

Required tools:

- Screwdriver: Hexagon-head size 5
or
- Screwdriver: Pozidriv PZ 2



► Remove the screws.



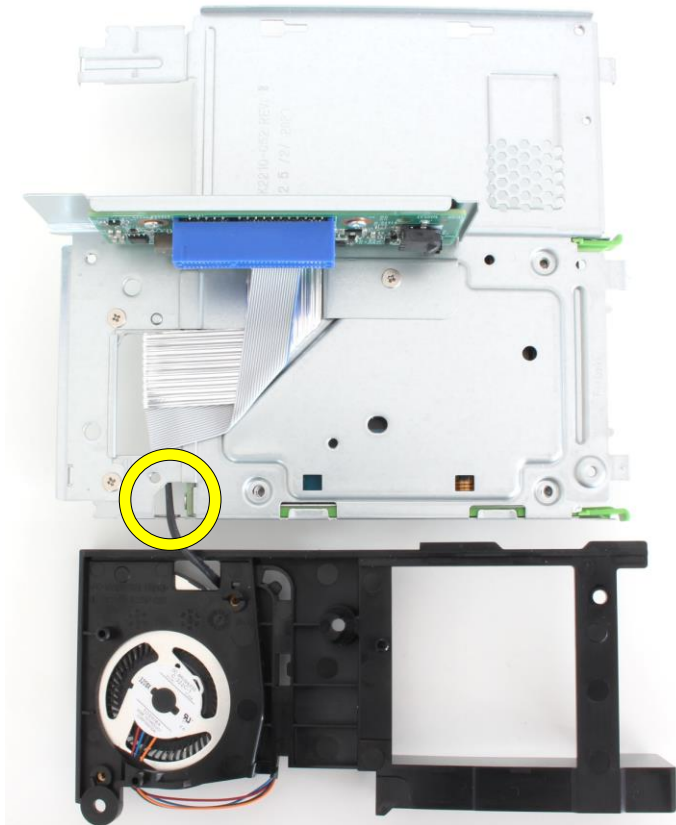
► Place the drive bay next to the unit and pull the fan cable out of the opening.

PCIe fan module:



Installation:

- ▶ Proceed in reverse order.



- ▶ Guide the fan cable through the marked opening when installing.

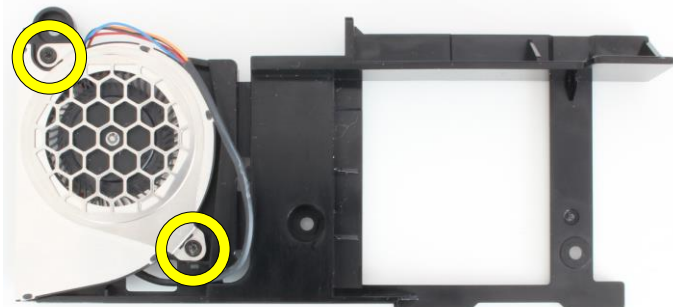
Remove the PCIe fan (Housing internal PSU)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the PCIe fan module \(Housing internal PSU\)](#)

Required tools:

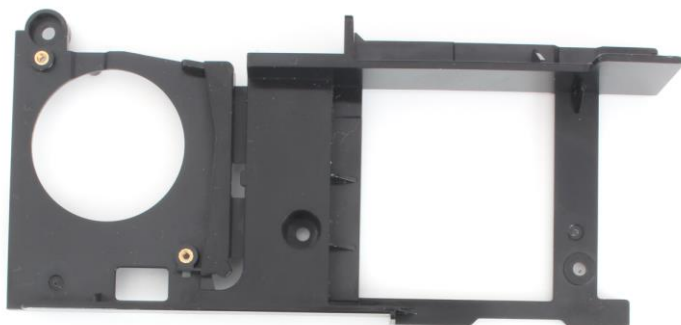
Screwdriver: Torx Plux 6IP



- ▶ Remove the screws and remove the fan.



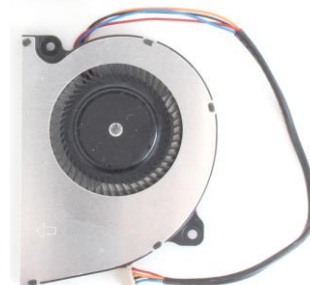
Fan support:



Fan grille:

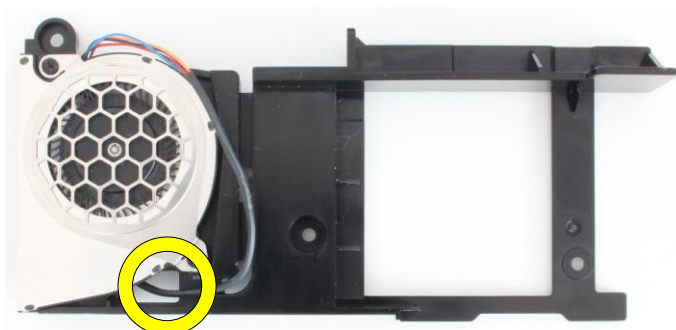


Fans:



Installation:

- ▶ Proceed in reverse order.



- ▶ Guide the fan cable through the marked opening when installing.

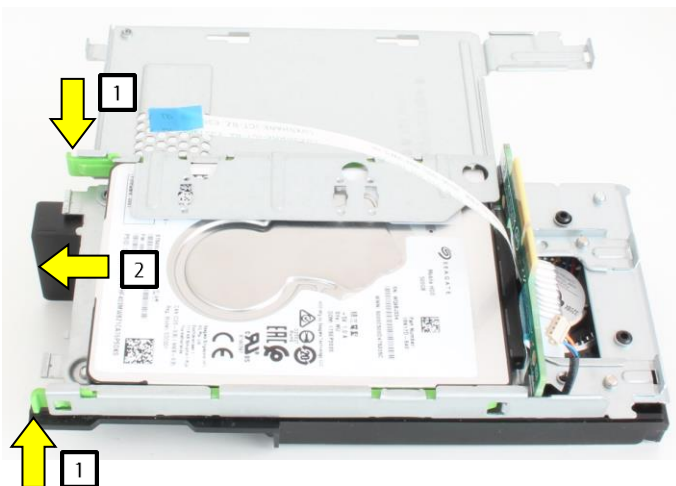
Remove the 2.5-inch drive (Housing internal PSU)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)

Required tools:

- None



- Press the rails together (1) and disconnect the drive in the direction of the arrow (2).



- Remove the hard disk rails (1) and disconnect the cable (2).

Installation:



- ▶ Attach the rails to the drive (insert pins into the threaded holes). Pay attention to the direction.
- ▶ Slide the drive into its bay until it snaps in place.

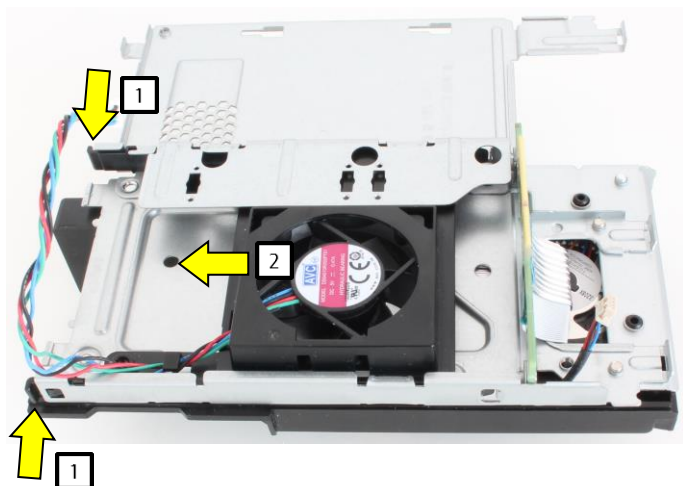
Remove the memory cooling fan (Housing internal PSU)

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)

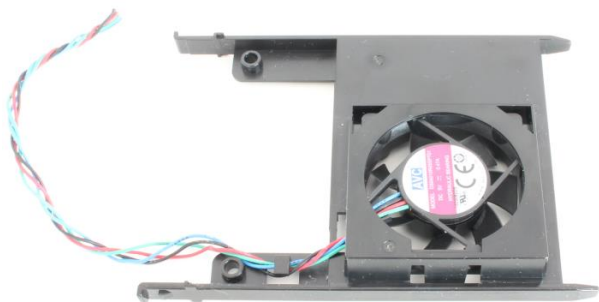
Required tools:

- None



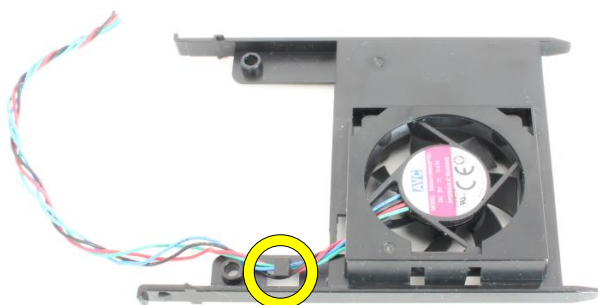
- ▶ Press the rails together (1) and disconnect the drive in the direction of the arrow (2).

Memory cooling fan:



Installation:

- ▶ Proceed in reverse order.



- ▶ Make sure that the fan cable is in the cable guide.



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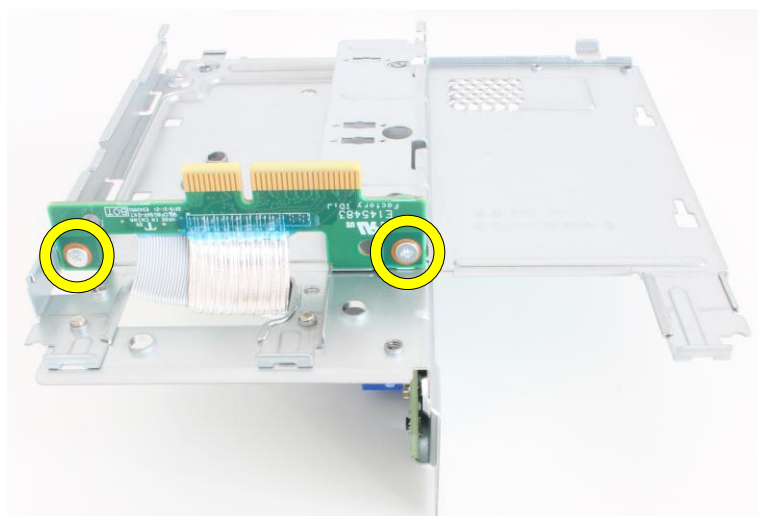
Remove the Riser board (Housing internal PSU)

Required work steps:

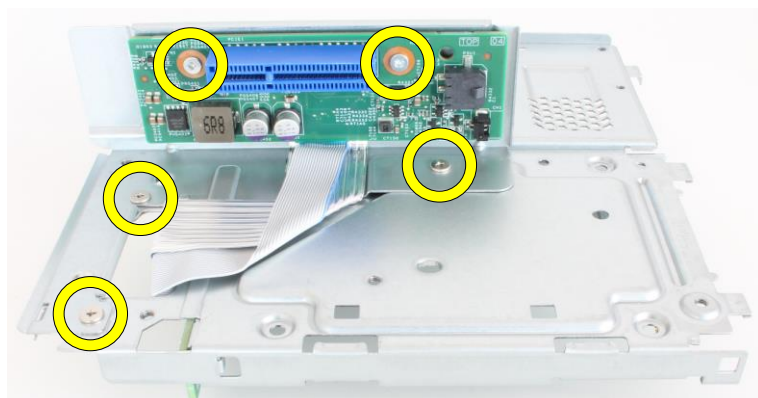
- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the PCIe fan module \(Housing internal PSU\)](#)
- [Remove the 2.5-inch drive \(Housing internal PSU\)](#)
- [Remove the memory cooling fan \(Housing internal PSU\)](#)

Required tools:

- Screwdriver: Phillips PH 1

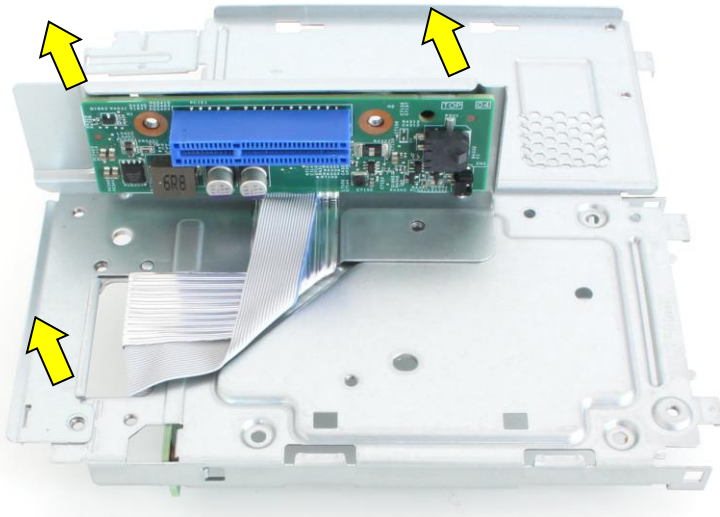


► Remove the screws.

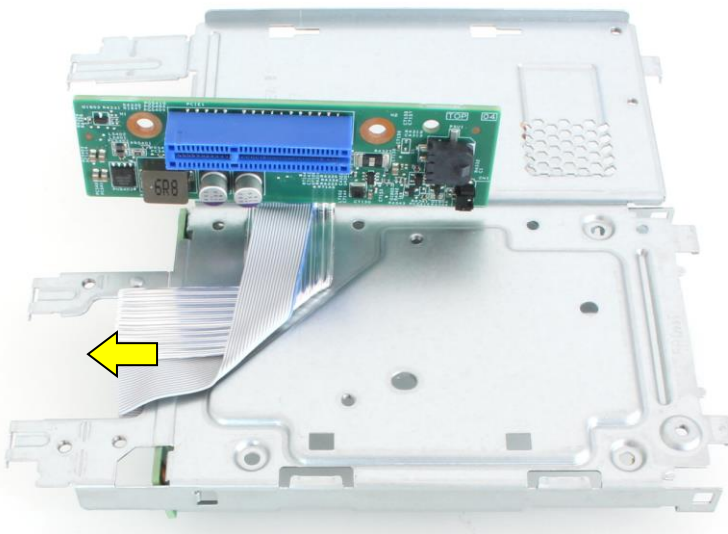


► Remove the screws.





► Remove the metal bracket.

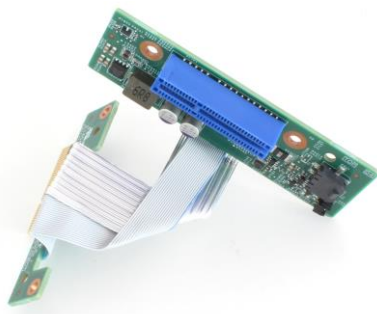


► Remove the Riser board.

Riser board

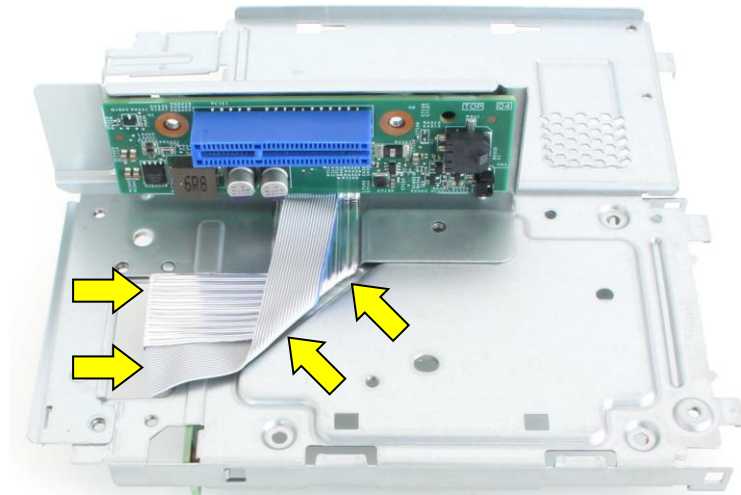
Metal bracket

Drive bay



Installation:

- ▶ Proceed in reverse order.



- ▶ When installing, make sure that the cable is folded correctly.

Remove the I/O panel

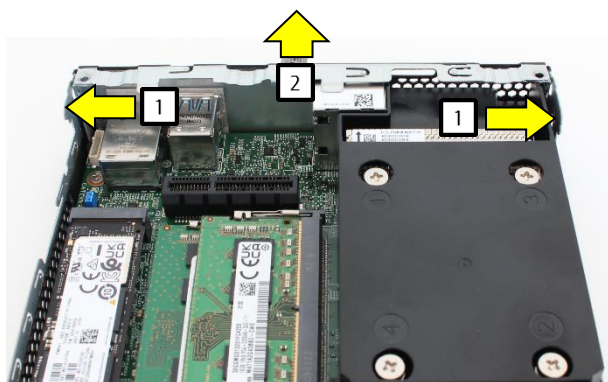
Required work steps:

- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)

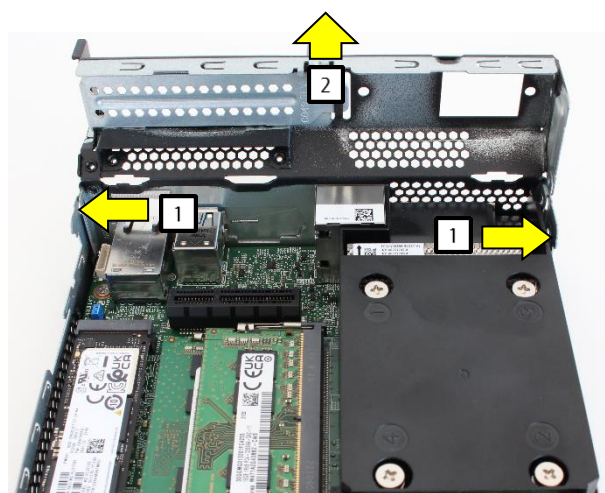
Required tools:

- Screwdriver: Phillips PH 1

Housing small and Housing ODD

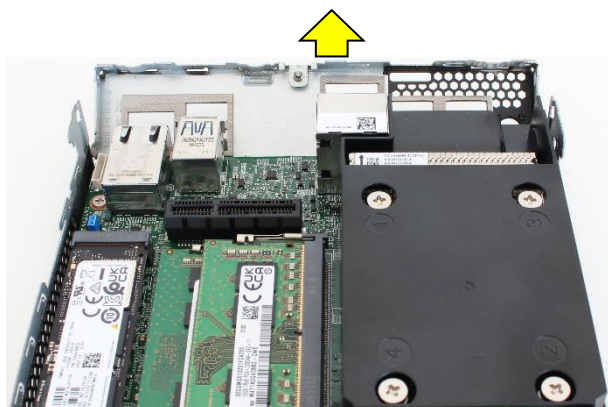


Housing internal PSU

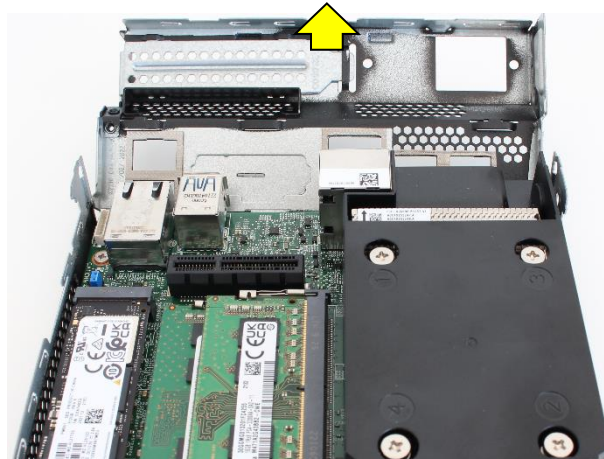


- ▶ Push the side panel slightly apart (1) and tilt the panel backwards (2).

Housing small and Housing ODD



Housing internal PSU



- ▶ Remove the cover.

Installation:

- ▶ Proceed in reverse order.



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Remove the speaker

Required work steps:

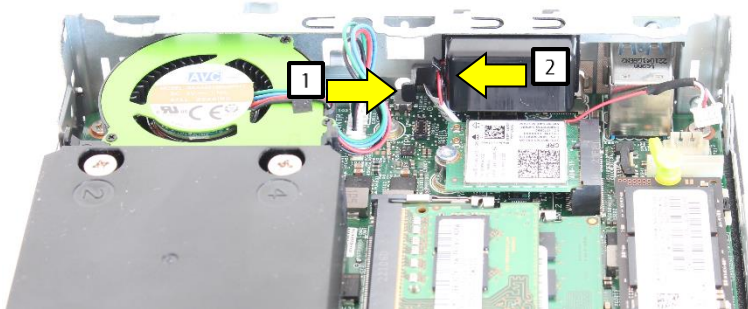
- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)

Required tools:

- None



- ▶ Unplug the cable from the mainboard.



- ▶ Press the plastic latch in the direction of the arrow (1) to open the bracket. Slide the speaker in the direction of the arrow (2).
- ▶ Remove the speaker.

Loudspeaker:



Installation:

- ▶ Proceed in reverse order.



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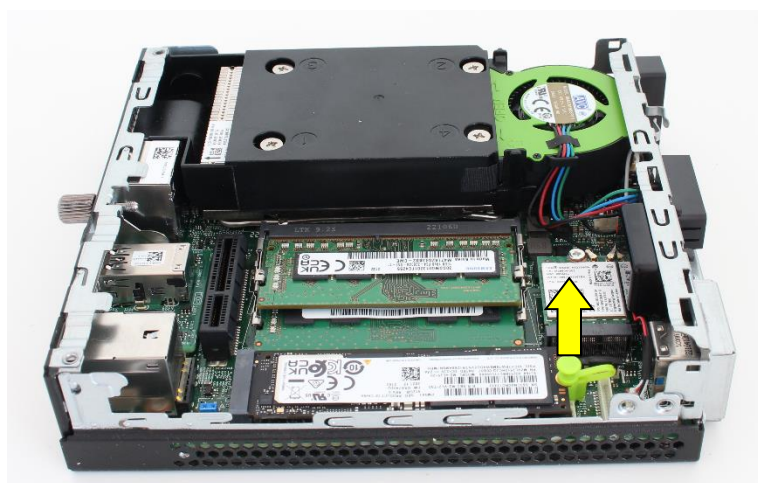
Remove the M.2 SSD module

Required work steps:

- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)

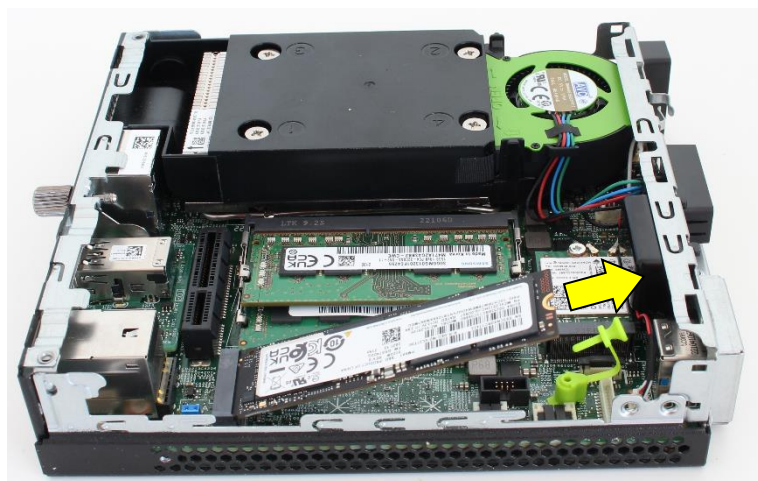
Required tools:

- None



S

- Remove the pin from the connector lock.
The module will lift up.



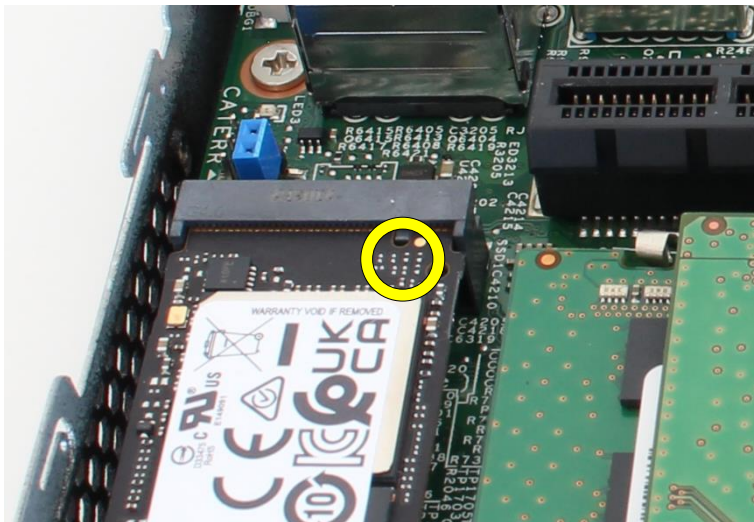
- Disconnect the module.

System without M.2 SSD module:



Installation:

- ▶ Proceed in reverse order.



- ▶ Make sure to align it properly when inserting the module.

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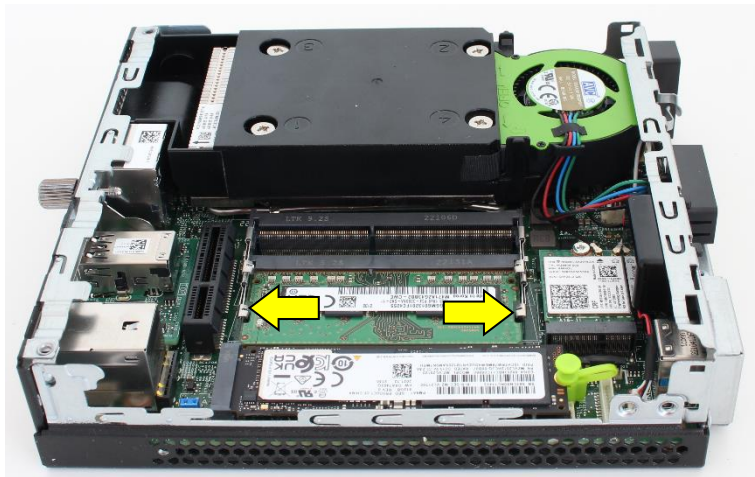
Remove memory modules

Required work steps:

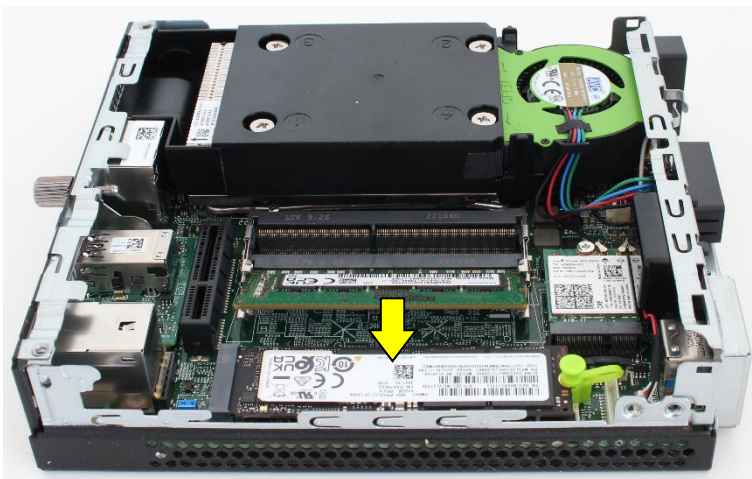
- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)

Required tools:

- None



- ▶ Release the memory module by pulling the retaining clips apart. The module will lift up.



- ▶ Disconnect the module.

This image shows the internal components of a Dell R710 server chassis. The motherboard is populated with two 8GB DDR3 ECC RAM modules in the DIMM slots. A green fan is mounted on the right side of the chassis. The front panel of the chassis is visible at the bottom, showing the perforated metal grille. The server is oriented vertically, with the front panel on the left and the back panel on the right.

- ▶ Proceed in reverse order.



- ▶ Make sure to align it properly when inserting the module.

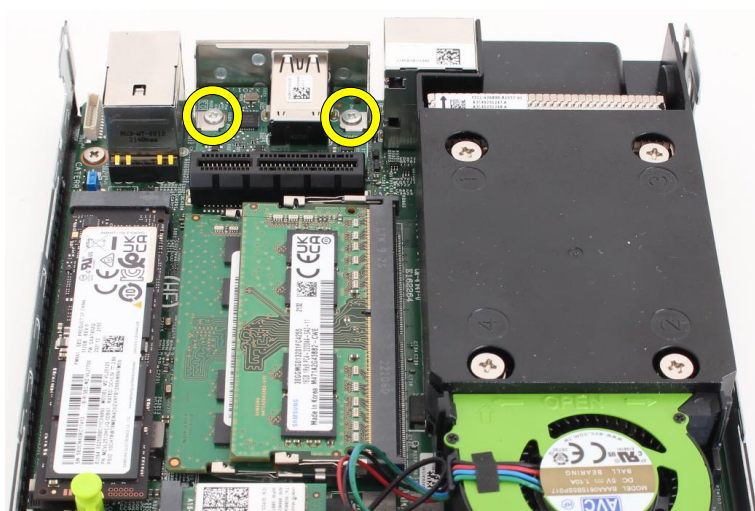
Remove the Flexible Interface Option

Required work steps:

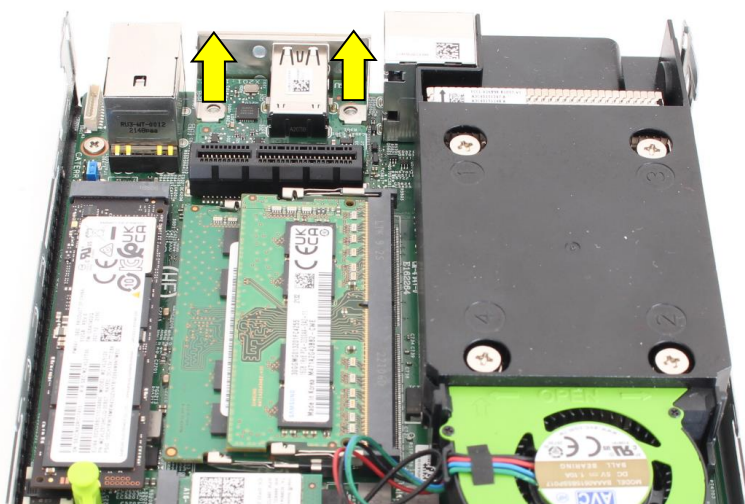
- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the I/O panel](#)

Required tools:

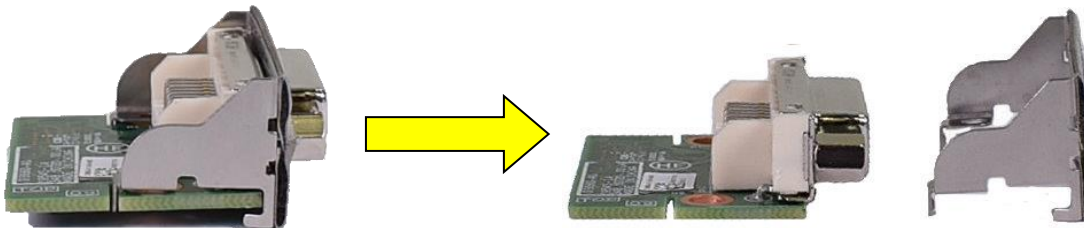
- Screwdriver: Phillips PH 1



- Remove the screws.



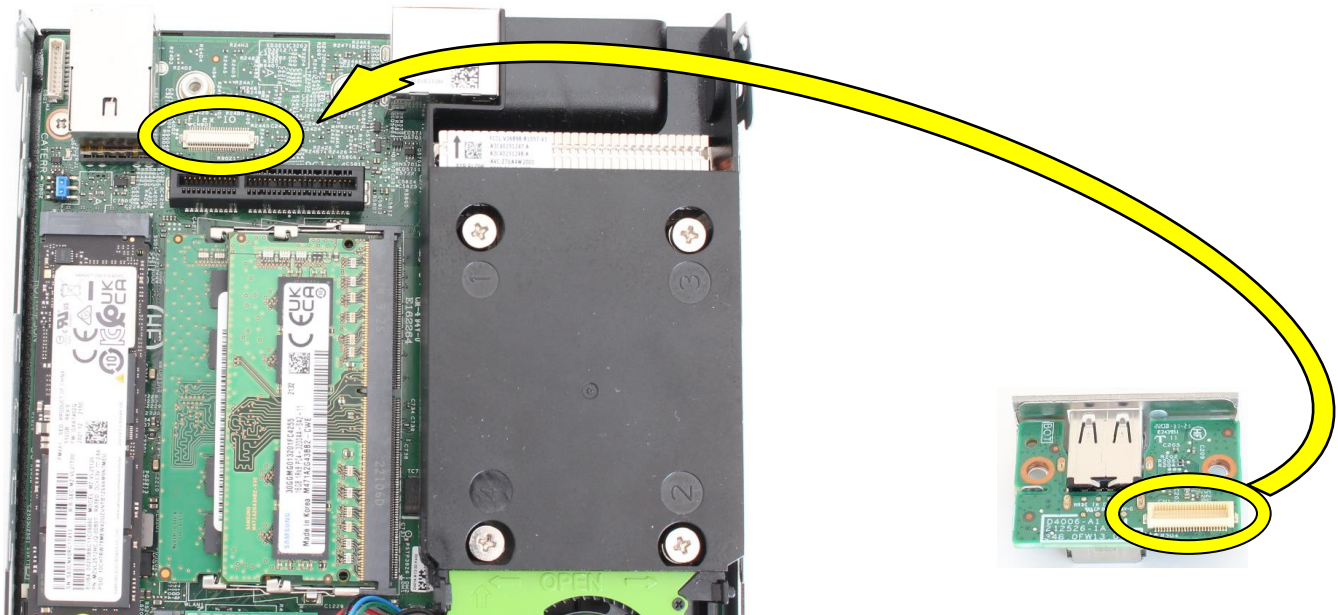
- Lift the Flexible Interface Option upwards in order to remove it from its slot in the mainboard.



- ▶ Remove the Flexible Interface Option from the metal frame.

Installation:

- ▶ Proceed in reverse order.
- ▶ Make sure to properly fit the interface into the metal frame (see above).



- ▶ Make sure to insert the Flexible Interface Option into its slot on the mainboard.

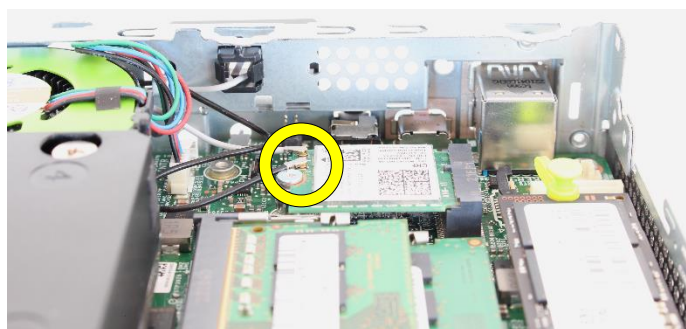
Remove the M.2 Wi-Fi module

Required work steps:

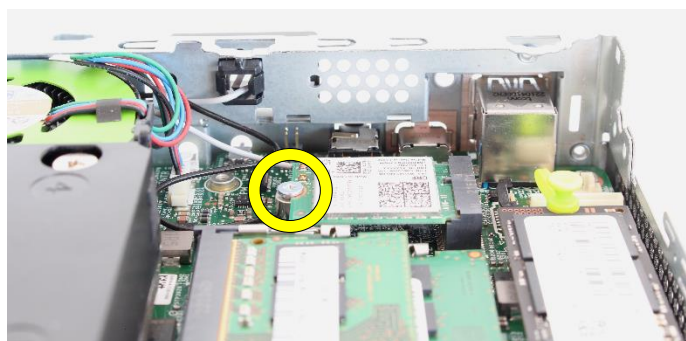
- [Open the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the speaker](#)

Required tools:

- Screwdriver: Phillips PH 1



- ▶ Remove the antenna cable from the module.



- ▶ Remove the screw.



- ▶ Disconnect the module.

Installation:

- ▶ Proceed in reverse order.

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Binding Support Bulletins for disassembly



[SB-PC-10016](#)

Minimum requirements for the proper completion of hardware repairs on deskbound systems



[SB-PC-11009](#)

Damage to the mainboard due to improper handling during repairs



[SB-PC-11001](#)

Information about mainboard (spare parts) replacement with LGA115x, LGA1366 and LGA2011-x sockets



[SB-PC-15002](#)

WIN8x & WIN10 - Replacement product key for repaired devices



For any information about ESD, please refer to the web-based training course "[ESD in electronics](#)"

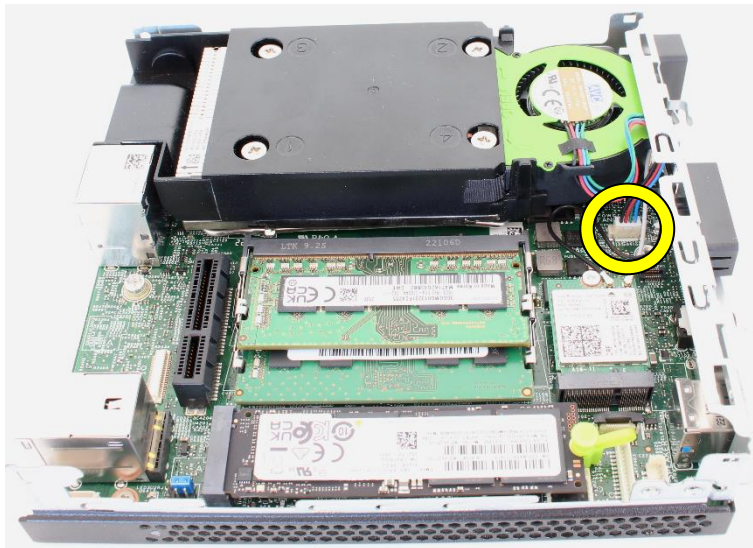
Remove the heat sink

Required work steps:

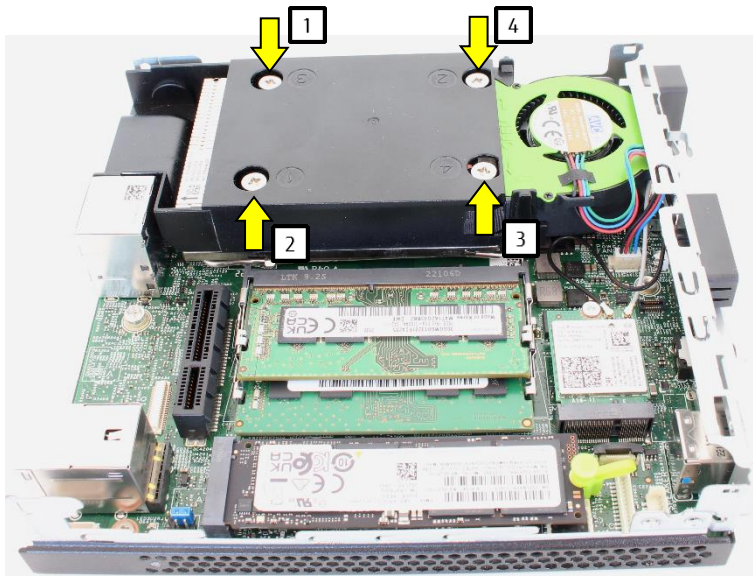
- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the I/O panel](#)

Required tools:

- Screwdriver: Phillips PH 1



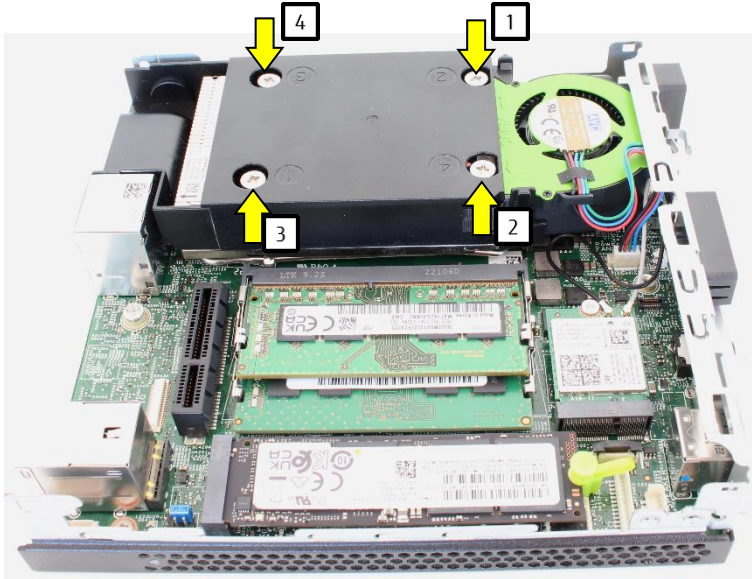
- Disconnect the fan cable from the mainboard.



Do not loosen the screws crosswise, instead loosen them one by one from 1 to 4 until they are completely unscrewed.

- Remove the heat sink from the system board.

Installation:



- Proceed in reverse order.

Remember to replace the missing thermal grease.

It may also be necessary to remove any dried thermal grease on the top on the CPU and on the bottom on the heatsink before reapplying thermal grease.

Remove the CPU

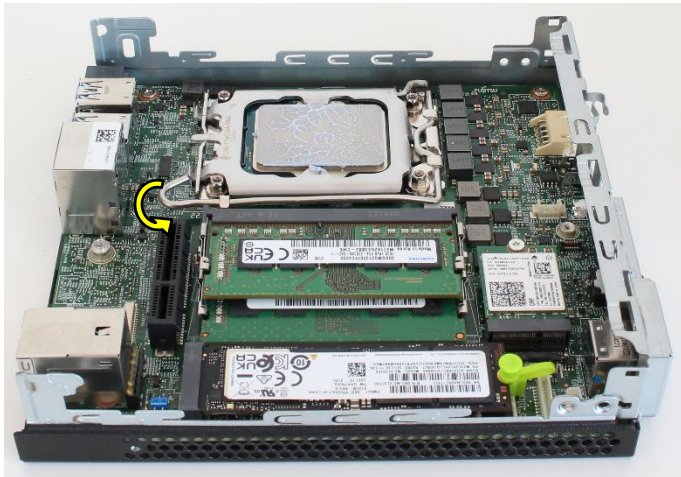
Please refer to the Support Bulletin [SB-PC-11001](#) ("Information concerning exchange of mainboard (spares) with LGA115x or LGA1366 sockets")!

Required work steps:

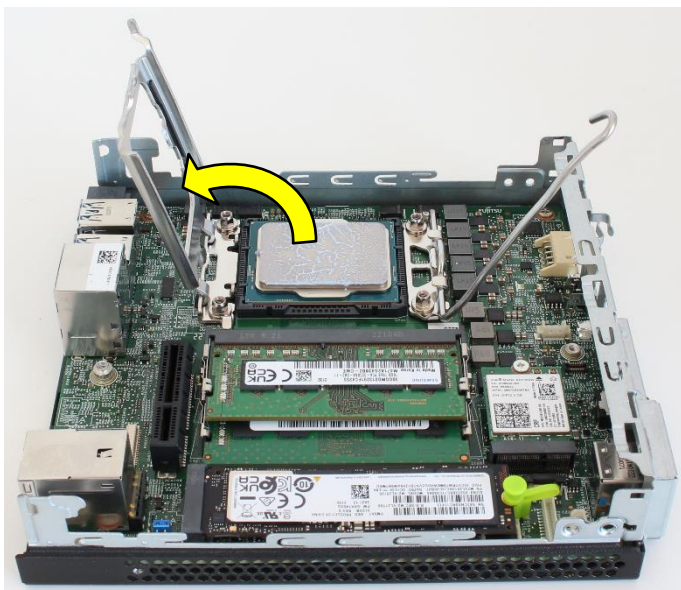
- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the I/O panel](#)
- [Remove the heat sink](#)

Required tools:

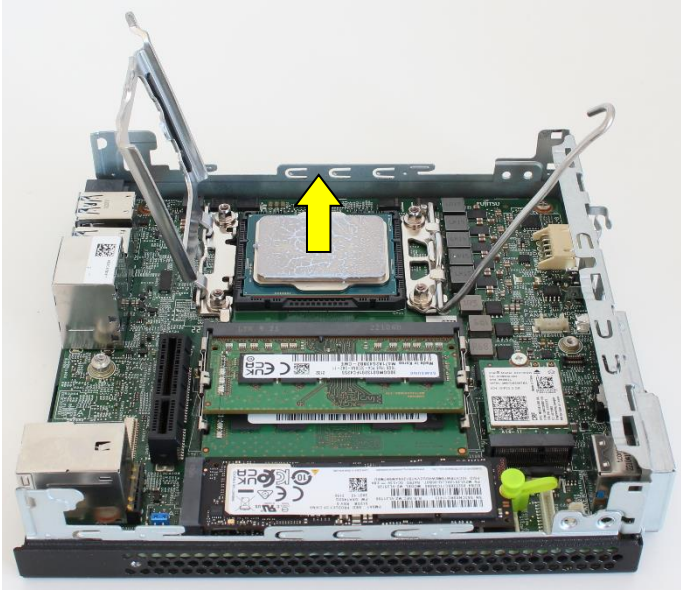
- None



- ▶ Release the lever on the CPU bracket by sliding it slightly down and then sideways.



- ▶ Open the CPU bracket.



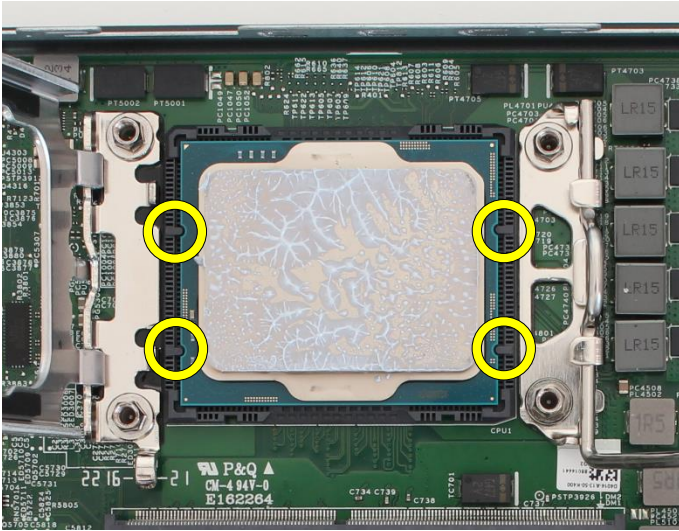
- Remove the CPU.

System view without CPU:



- If you want to return the mainboard, put the protective cover on the CPU base before closing the CPU bracket.

Installation:



- ▶ Proceed in reverse order.
- ▶ Pay attention to the CPU coding (see [Support Bulletin SB-PC-11001](#)).
Remember to replace the missing thermal grease.
It may also be necessary to remove any dried thermal grease on the CPU and on the heatsink before reapplying thermal grease.

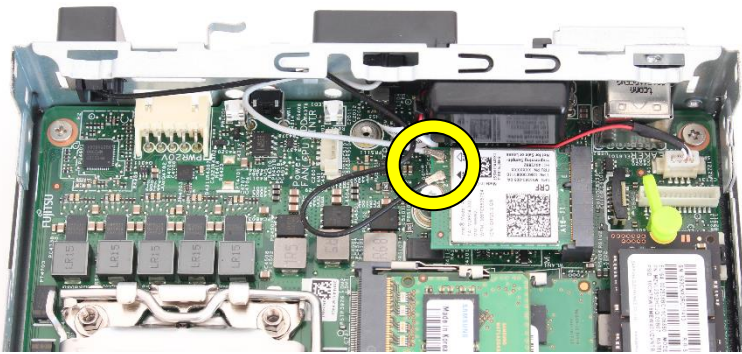
Remove the Wi-Fi antennas

Required work steps:

- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the I/O panel](#)
- [Remove the heat sink](#)

Required tools:

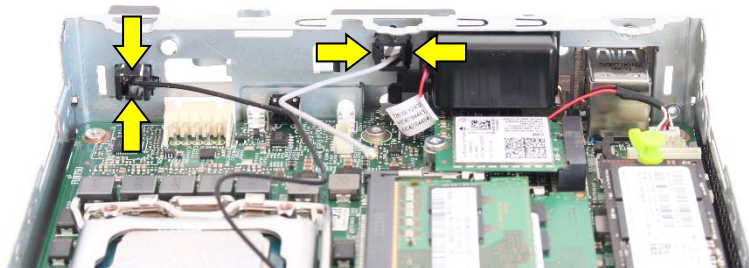
- None



- ▶ Remove the antenna cable from the module.



- ▶ Unhook the cable from the bracket.



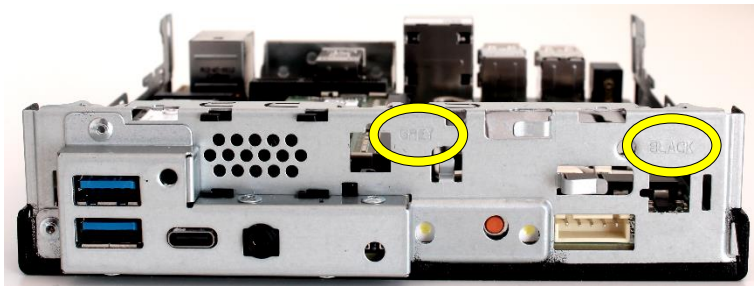
- ▶ Press and hold the antennas' plastic latches in place.
- ▶ Press the bracket outwards.

Antennas:



Installation:

- Proceed in reverse order.



- Feed the cables through the openings (grey cable > GREY/black cable > BLACK).



- Ensure proper alignment when fitting the brackets into the cable guides of the casing.



- Pay attention to the colour coding.

Remove the mainboard

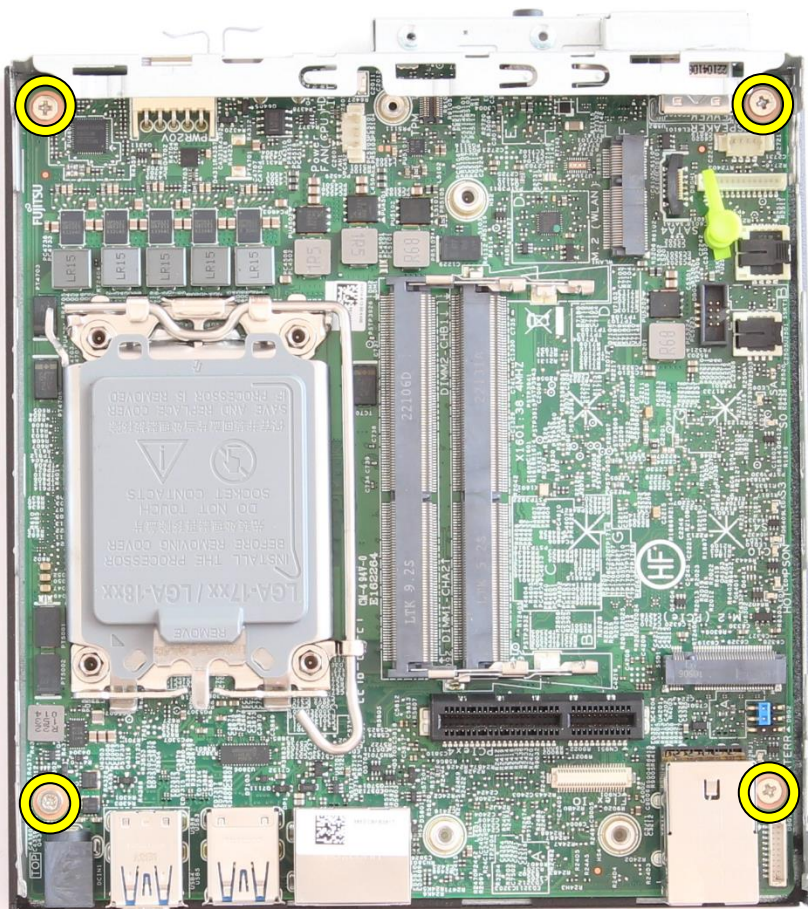
Please refer to the Support Bulletin [SB-PC-11001](#) ("Information concerning exchange of mainboard (spares) with LGA115x or LGA1366 sockets")!

Required work steps:

- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the I/O panel](#)
- [Remove the speaker](#)
- [Remove the M.2 SSD module](#)
- [Remove memory modules](#)
- [Remove the Flexible Interface Option](#)
- [Remove the M.2 Wi-Fi module](#)
- [Remove the heat sink](#)
- [Remove the CPU](#)

Required tools:

- Screwdriver: Phillips PH 1



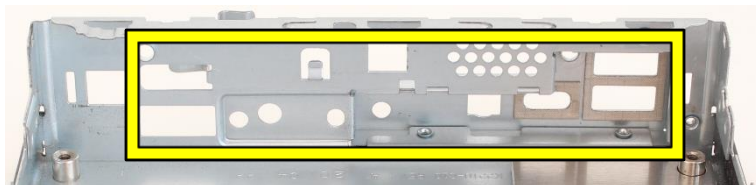
- Remove the screws.



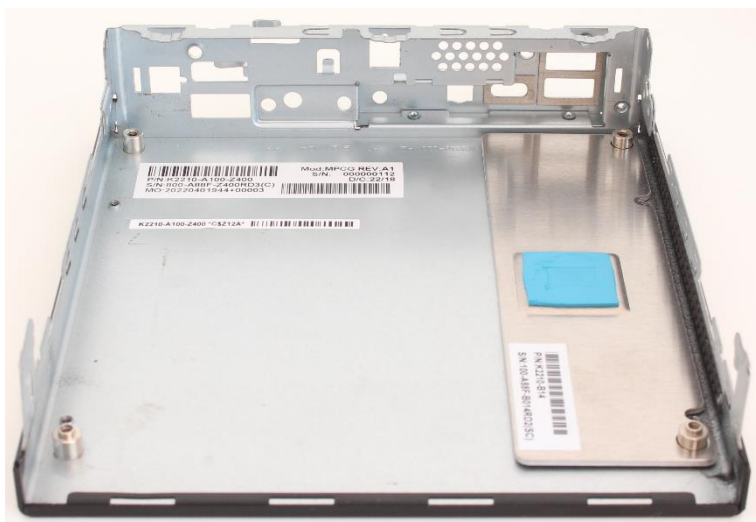
- ▶ Start by lifting the mainboard.
Remove the mainboard.

Installation:

- ▶ Proceed in reverse order.



- ▶ If you are fitting the mainboard into the casing, first route the I/O connectors through the corresponding openings before inserting the mainboard.



BoardLock



- ▶ The final position of the mainboard is determined by the BoardLock connectors.

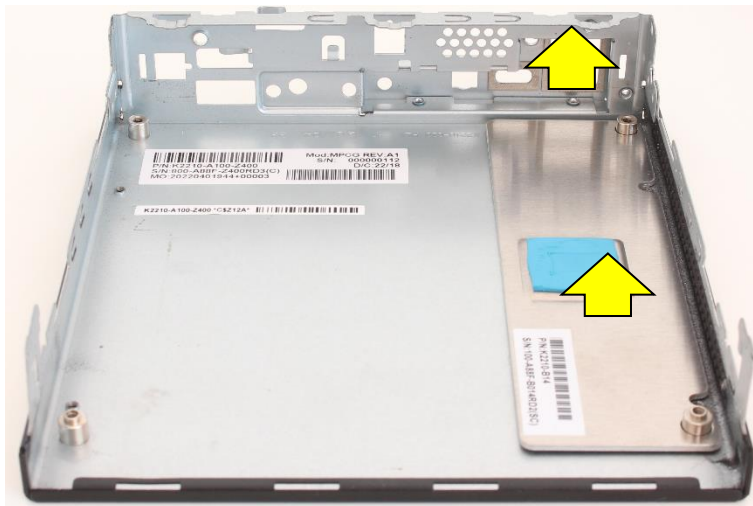
Remove the cooling plate

Required work steps:

- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the I/O panel](#)
- [Remove the mainboard](#)

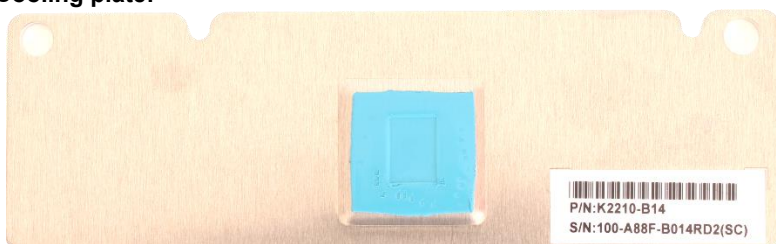
Required tools:

- None



- Lift the cooling plate out.

Cooling plate:



Installation:

- Proceed in reverse order.

[↑](#) return to Table of Contents

Remove CMOS-battery

Required work steps:

- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the I/O panel](#)
- [Remove the mainboard](#)

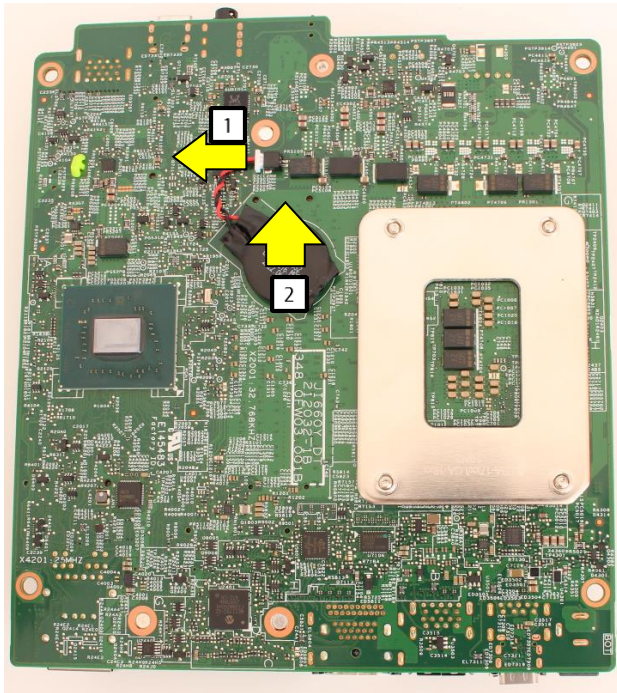
Required tools:

- None



The improper exchanging of the lithium battery can result in a danger of explosions!
The lithium battery may be replaced only by a battery recommended by the manufacturer.
Do not dispose of lithium batteries in the household waste. They must be disposed of in accordance with the local regulations, as hazardous waste

Back of the mainboard:



- ▶ Disconnect the plug (1) and remove the battery from the mainboard (2).

Battery:



Installation:

- ▶ Proceed in reverse order.



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Remove the pluggable Trusted Platform module

Required work steps:

- [Open the cable cover](#)
- [Remove the cable cover](#)
- [Remove the pedestal](#)
- [Remove the dust filter](#)
- [Open the case](#)
- [Remove the drive bay \(Housing small\)](#)
- [Remove the drive bay \(Housing ODD\)](#)
- [Remove the PCIe card \(Housing internal PSU\)](#)
- [Remove the power supply unit \(Housing internal PSU\)](#)
- [Remove the drive bay \(Housing internal PSU\)](#)
- [Remove the I/O panel](#)
- [Remove the mainboard](#)

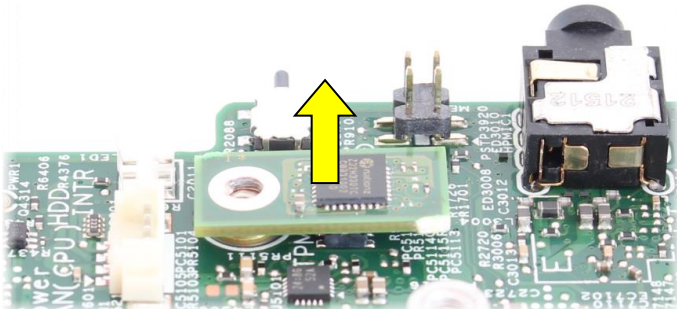
Required tools:

- Screwdriver: Y-Type Y1

Trusted Platform module on the mainboard:

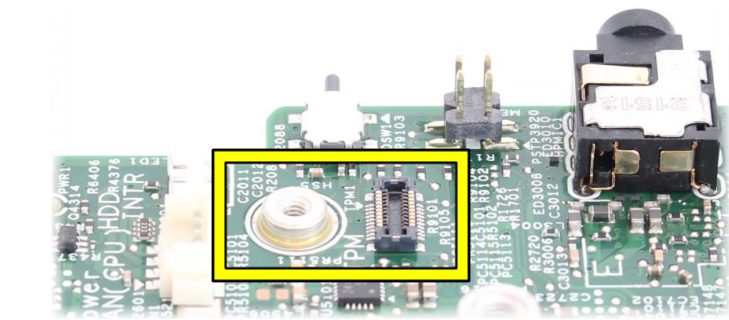


- ▶ Remove the screw.

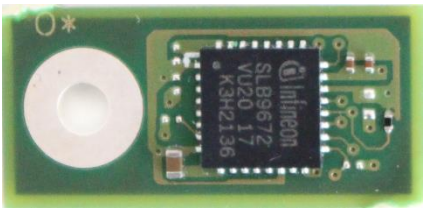


- ▶ Remove the module.

TPM slot:



TPM module:



Installation:

- ▶ Proceed in reverse order.



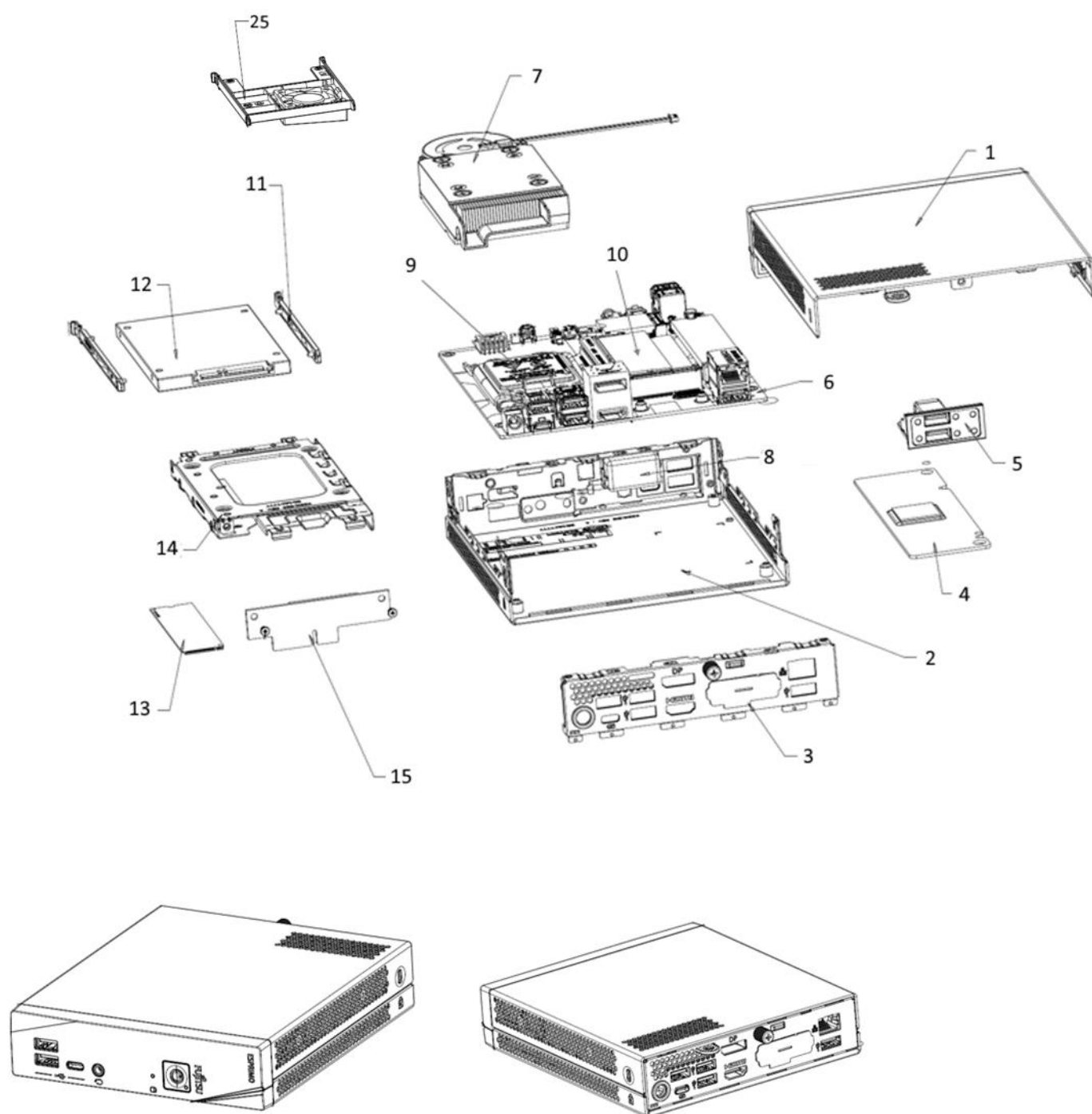
- ▶ Secure the module.



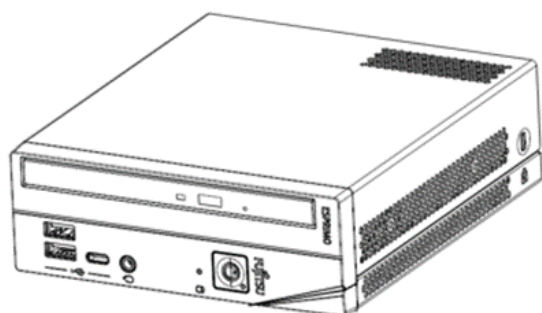
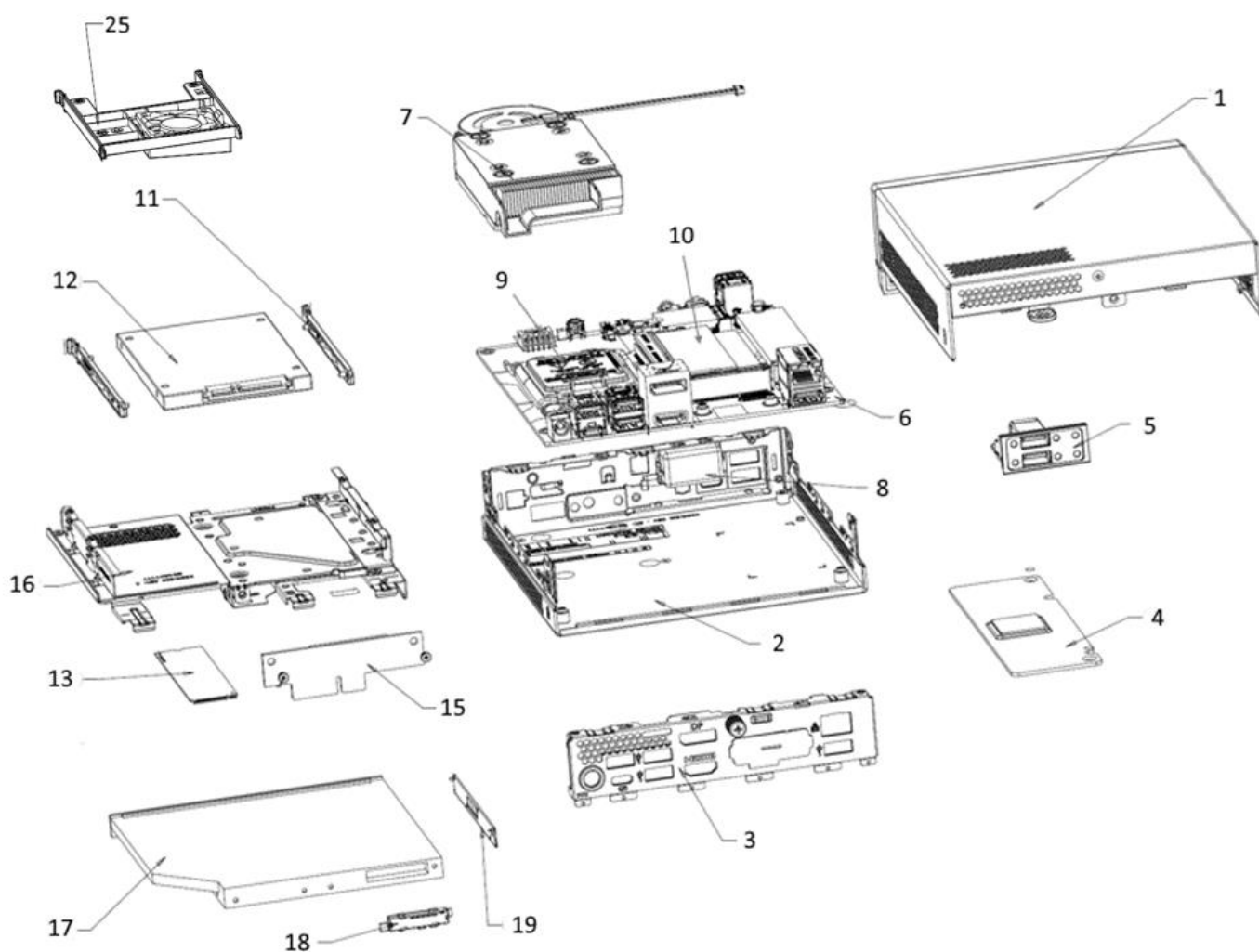
Spare parts ESPRIMO G6012 / G9012 / G9013

Pos.	Spare part	Internal designation	Replaceable parts		Upgradeable	Repairable
			Customer	Service Partner		
1	Housing cover	Top Cover	X			
2	Bottom part of the housing	Bottom Case	X			
3	I/O panel	Rear I/O Bracket	X			
4	Cooling plate	Cooling Plate	X			
5	Flexible Interface Option	Flex I/O	X		X	
6	Mainboard	depending on the device configuration		X		X
7	Heat sink	Heatsink		X		
8	Loudspeaker	Speaker	X			
9	Processor	Depending on the device configuration		X	X	
10	Memory modules	Depending on the device configuration	X		X	
11	Drive rails	EasyRail	X		X	
12	2,5" HDD/SDD	Depending on the device configuration	X		X	
13	M.2 SSD module	Depending on the device configuration	X		X	
14	Drive bay	HDD Carrier	X		X	
15	Riser board	M.2 Riser	X		X	
16	Drive bay	ODD Carrier	X		X	
17	Optical drive	ODD Drive	X		X	
18	Optical drive cable	ODD Cable	X		X	
19	Drive rail	ODD Easy Rail	X		X	
20	HDD/SSD cable	HDD Cable	X		X	
21	Drive bay	Device Carrier	X			
22	Riser board cable	Riser Cable	X		X	
23	Low profile PCIe board	Depending on the device configuration	X		X	
24	Power supply unit	PSU	X			
25	Memory cooling fan	DDR5 AXIAL FAN	X			
26	Fan support	NB FAN HOLDER	X		X	
27	PCIe fan	NB FAN	X		X	
28	Fan grille	FAN GRID	X		X	
without illustration	Battery	Battery		X		
without illustration	Power supply unit	AC Adapter	X			
without illustration	Antennas	ANTENNA		X		

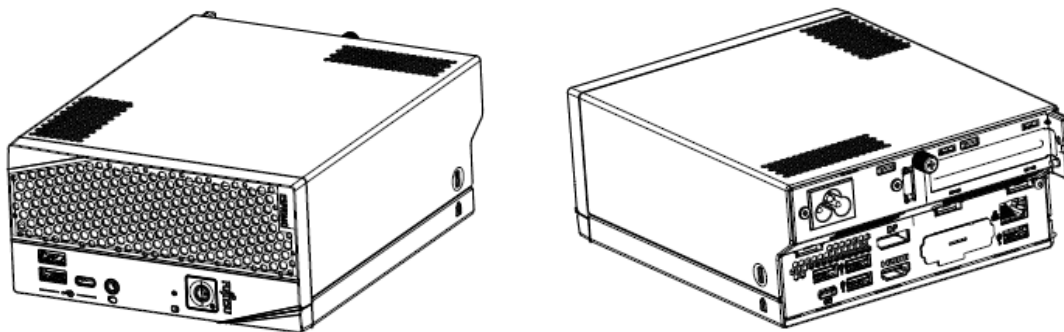
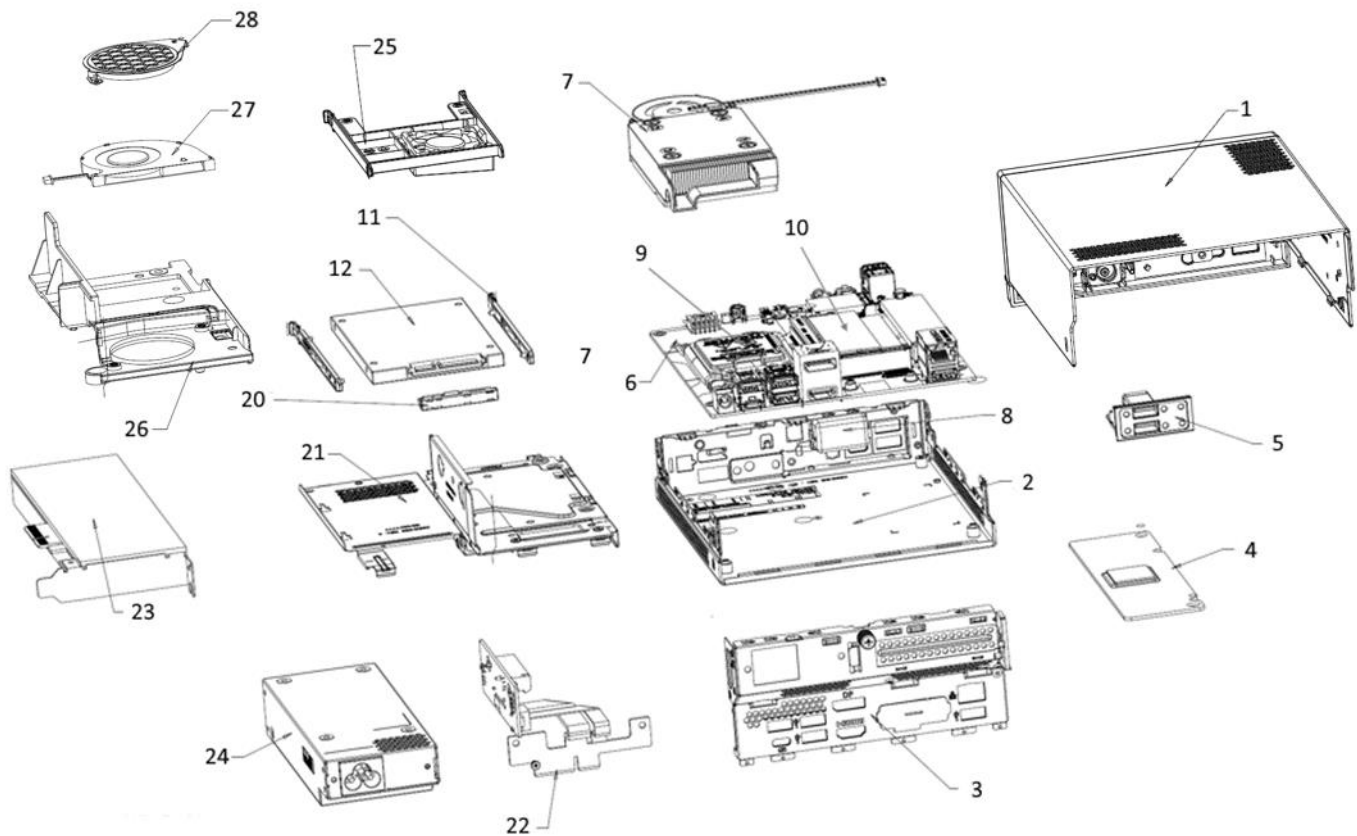
ESPRIMO G6012 / G9012 / G9013 (Housing small)



ESPRIMO G6012 / G9012 / G9013 (Housing ODD)



ESPRIMO G9012 / G9013 – (Housing internal PSU)



■ Spare part availability: 5 years

Spare parts can be obtained from our approved and certified service partners:

[Fujitsu Technical Support pages from Fujitsu EMEA](#)

Disposal information for recycling companies

These instructions are for the complete dismantling of a Fujitsu product. All materials can thereby be disposed of according to their provision in the EC directive 2012/19/EC (Waste Electrical and Electronic Equipment (WEEE)).

Product group: Desktop

Models: ESPRIMO G6012 / G9012 / G9013

Materials which must be disposed of in a special manner		
Designation	Remarks	Quantity
Printed circuit boards With a surface area greater than 10 cm ²	Mainboard	1
	Depends on the system configuration:	
	Memories	1-2
	WLAN module	0-1
	M.2 Module	0-2
	SSD	0-1
	HDD	0-1
	ODD	0-1
	Flexiboard	0-1
	Depending on the system/configuration, one of the following power supplies (PSU) is used in the system. Please check the version on the power supply.	
	Power supply S26113-E660-V50	1
Batteries	Lithium button batteries CR2032 (included on the mainboard)	1
Components containing mercury	Not included	
Liquid crystal displays (if applicable, together with the casing) with a surface area greater than 100 square centimetres and backlit displays with gas discharge lamps	Not included	
Cathode ray tubes	Not included	
Capacitors containing PCBs (PolyChlorinated Biphenyls)	Not included	

Materials which must be disposed of in a special manner		
Designation	Remarks	Quantity
Electrolytic capacitors which contain critical substances (height > 25 mm; diameter: > 25 mm or proportional similar volumes).	Depending on the system/configuration, one of the following power supplies (PSU) is used in the system. Please check the version on the power supply.	
	Power supply S26113-E660-V50	1
External electrical cables	Not permanently connected to the device	
Gas discharge lamps	Not included	
Plastics which contain brominated flame retardant	This product may contain plastic components containing brominated flame retardants. To the best of our knowledge, we provide this information for sorting them out. The position of the plastic components can be visually identified by their typical design function on the product (as shown in the disassembly manual). Many of these components (> 25 g) are bromine-free. Irrespective of this, these components are marked in accordance with ISO 11469 (usually marked directly on the plastic). A typical marking looks like this: > Polymer abbreviation - FR(xx) < i.e. > PC+ABS FR (40) < Code numbers for flame retardants (FR(xx)) can be found in ISO 1043-4. Codes for some brominated flame retardants: 14 Aliphatic/alicyclic brominated compounds 15 Aliphatic/alicyclic brominated compounds with antimony compounds 16 Aromatic brominated compounds (excluding brominated diphenyl ether and biphenyl) 17 Aromatic brominated compounds (other than brominated diphenyl ether and biphenyl) containing antimony compounds 22 Aliphatic/alicyclic chlorinated and brominated compounds 42 Brominated organic phosphorus compounds Plastic components weighing less than 25 grams may contain brominated flame retardants. These smaller plastic components are located in the entire PC system. These include, above all: Fans, fans in the power supply, speakers and components on PCIe cards.	
Chlorofluorocarbons (CFC), hydrochlorofluorocarbons (HCFC) or hydrofluorocarbons (HFC), hydrocarbons (HC)	Not included	
Toner cartridges, liquid and paste-like, and colour toner	Not included	
Asbestos waste and components which contain asbestos	Not included	
Components which contain fire-proof ceramic fibres	Not included	
Components which contain radioactive substances	Not included	